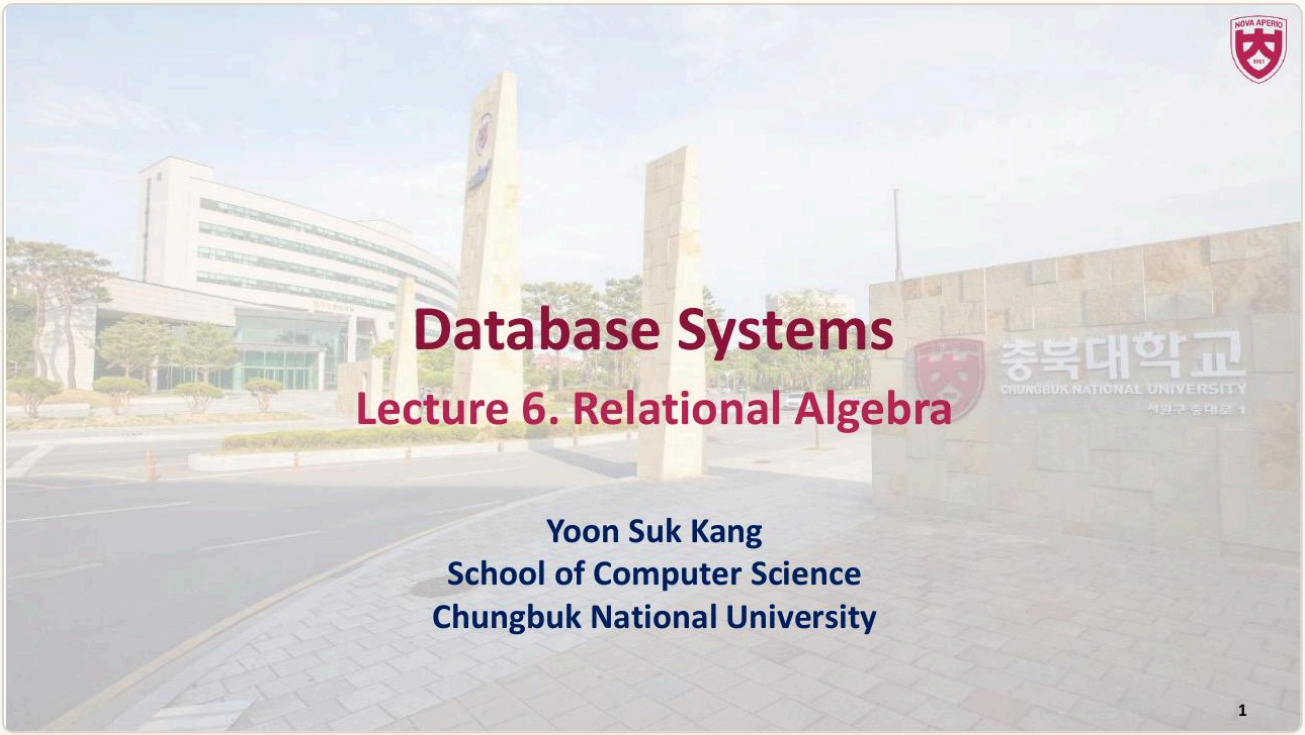
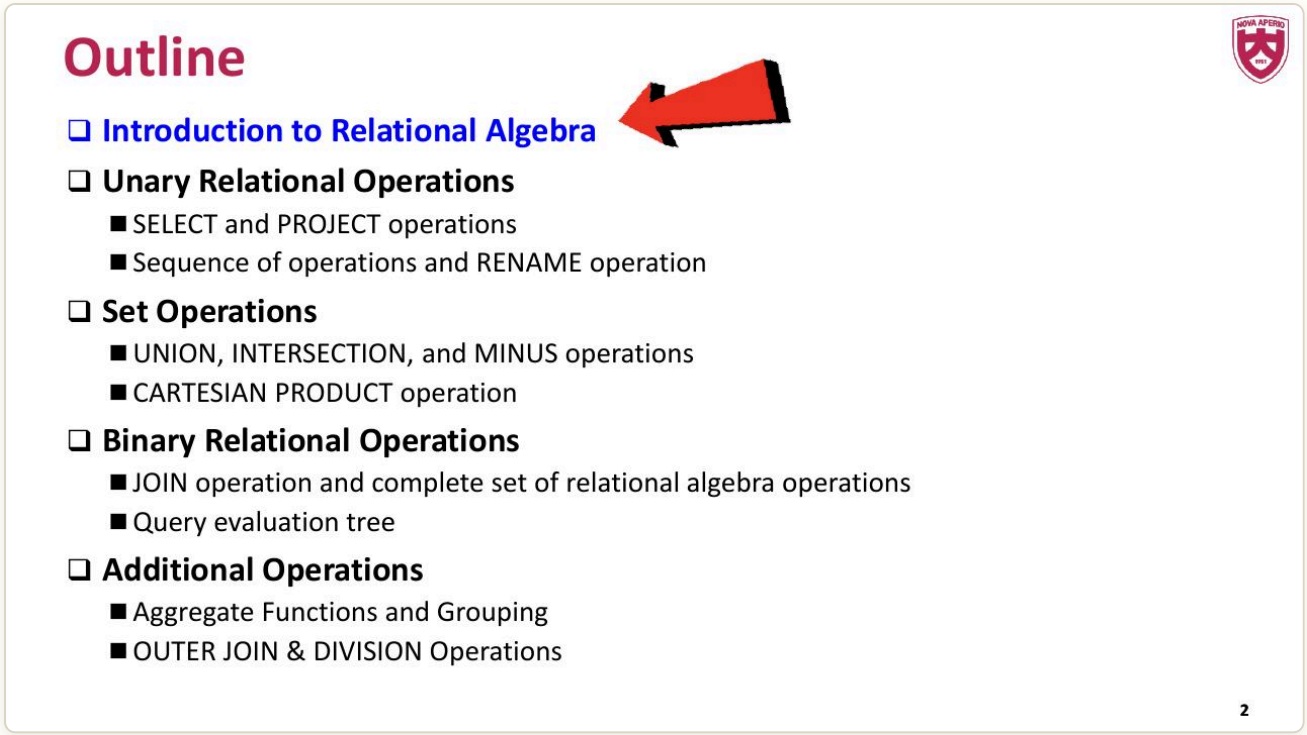


데이터베이스 시스템 — 6강: 관계 대수

- Yoon Suk Kang · 충북대 소프트웨어학부

✍️ 메모

💡 이해 기호 7개(σ π ρ ∪ ∩ − ×) + ⋈ * ⋉ ⋈ ÷ — 기호와 뜻을 짝으로

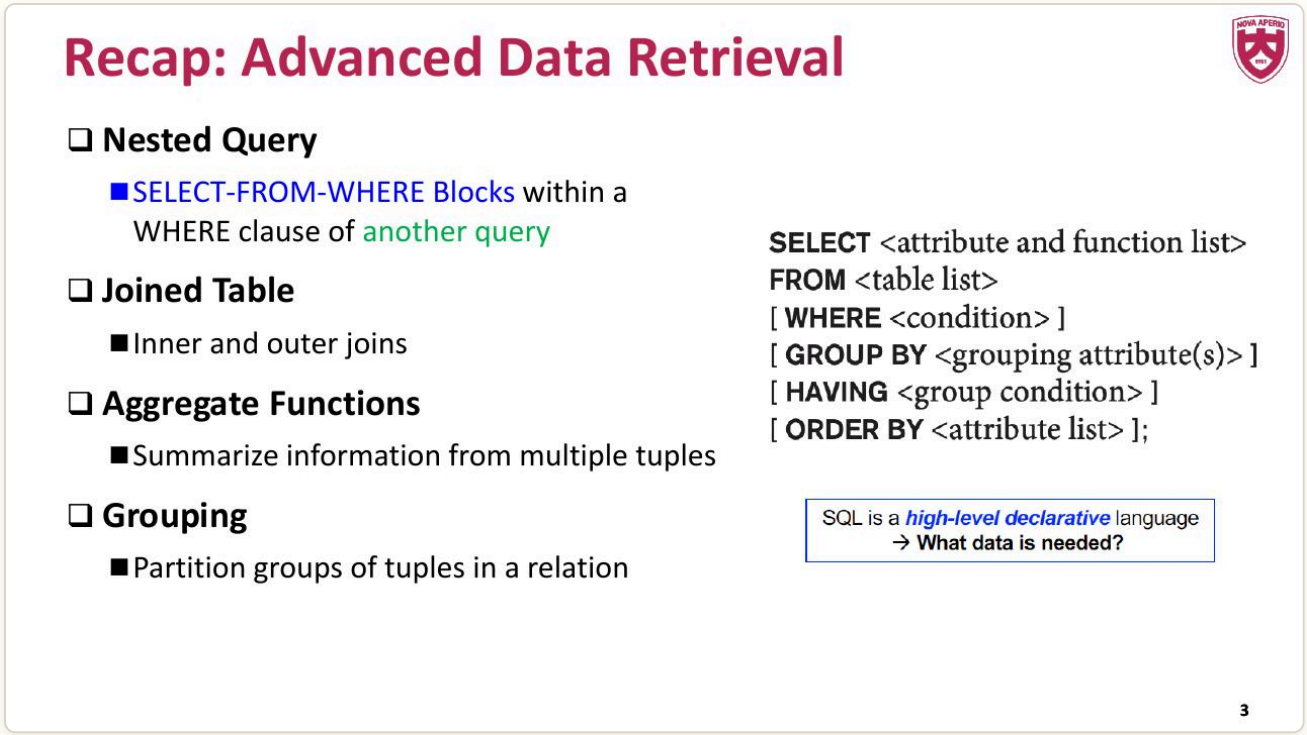


목차 (Outline)

- **관계 대수 소개**
- **단항 연산** — SELECT·PROJECT / 연산 잇기와 RENAME
- **집합 연산** — UNION·INTERSECTION·MINUS / CARTESIAN PRODUCT
- **이항 연산** — JOIN과 완전 집합 / 질의 평가 트리
- **추가 연산** — 집계 함수와 그룹화 / OUTER JOIN과 DIVISION

✍ 메모

— 넘어가도 되는 쪽



복습: 고급 데이터 검색 (4강)

- **Nested Query** — 다른 질의의 WHERE 안 SELECT-FROM-WHERE 블록
- **Joined Table** — inner·outer join
- **Aggregate Functions** — 여러 튜플의 정보를 요약
- **Grouping** — 튜플을 그룹으로 나눔

✍ 메모

— 넘어가도 되는 쪽

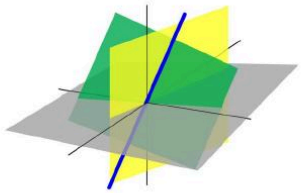
Introduction to Relational Algebra

❑ **What is the Algebra?**

- Represent numbers and quantities in equations with letters and symbols
- Basic operations: +, -, *, /

❑ **What is the Linear Algebra?**

- Linear transformation in vector spaces
- Basic operations
  - ❑ Vector addition and scalar multiplication, Matrix operations, …



❑ **What is the Relational Algebra?**

- Theoretical and mathematical model for "***how to get data***" (i.e., procedural)
- *The basic set of operations for the formal relational model*

4

관계 대수 소개

- 대수(Algebra): 방정식에서 수와 양을 문자·기호로 — +, −, *, /
- 선형대수: 벡터 공간의 선형 변환 — 벡터 덧셈, 스칼라 곱, 행렬 연산 …
- **관계 대수(Relational Algebra)**
  - '**어떻게** 데이터를 얻나'에 대한 이론적·수학적 모델 (**절차적, procedural**)
  - 형식적 관계 모델의 **기본 연산 집합**

✍ 메모

★ 시험 관계 대수 = procedural (어떻게) / SQL = declarative (무엇을)

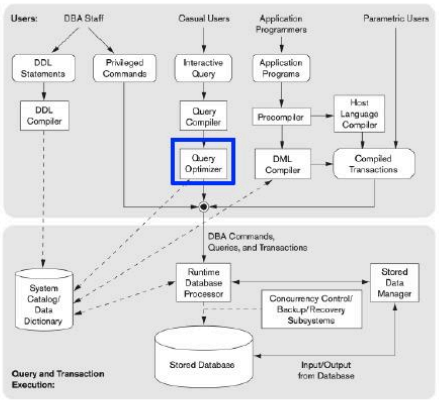
Motivation

❑ Why Relational Algebra Important?

- ■ Formal foundation for relational model operations
  - ❑ Selection, projection, renaming, join, cartesian product, …
  - ❑ Strictly based on the mathematical concepts of ***sets***
- ■ Core operations and functions in the internal modules are based on relational algebra operations
- ■ A basis for *implementing and optimizing queries in query processing and optimization* (Week 11)
  - ❑ *I.e., specifying a sequence of operations to get the results*

Thus, understanding relational algebra is crucial to define efficient query execution plans

Relational Algebra Operators $\pi \sigma \rho$ $\bowtie \cup \cap$



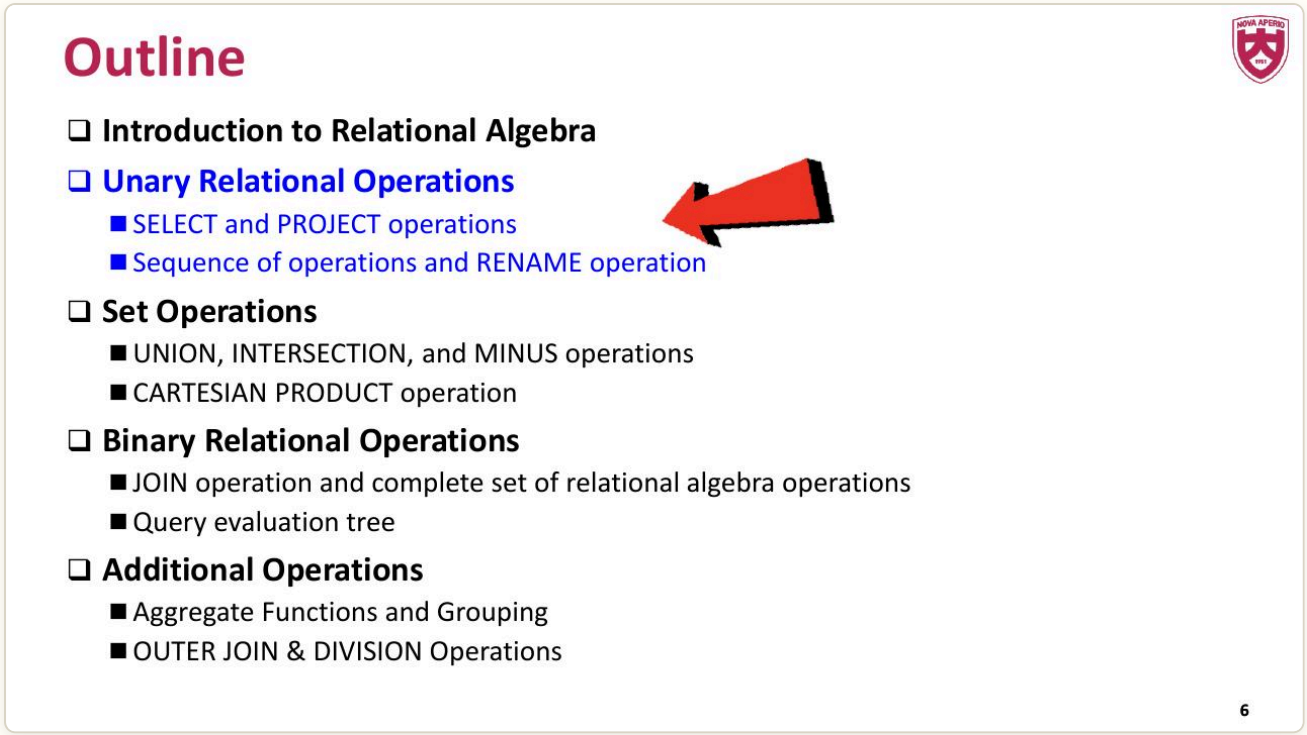
5

왜 중요한가?

- 관계 모델 연산의 **형식적 기초**
  - selection, projection, renaming, join, cartesian product …
  - **집합의 수학 개념**에 엄격히 기반
- DBMS 내부 모듈의 핵심 연산과 함수가 관계 대수 연산에 기반
- **질의 처리와 최적화**(11주)의 바탕
  - 결과를 얻는 **연산의 순서**를 지정

✍ 메모

— 넘어가도 되는 쪽



목차 (Outline)

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✍️ 메모

💡 이해 단항 연산부터

Unary Relational Operations: SELECT

❑ What is the SELECT Operation?

■ To choose a *subset of the tuples* from "**a relation**" satisfying a selection condition

❑ i.e., a **filter** that keeps only those tuples satisfying a qualifying condition, while discards the others

$$\sigma_{<\text{selection condition}>}(R)$$

■ Can be visualized as a *horizontal partition* into two sets of tuples

❑ Tuples satisfying the condition (selected) vs Tuples not satisfying the condition (filtered out)

$\sigma_{Dno=4}(EMPLOYEE)$

$\sigma_{Salary>30000}(EMPLOYEE)$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

The SELECT operation is **different** from the SELECT clause of SQL.

7

단항 연산: SELECT (σ)

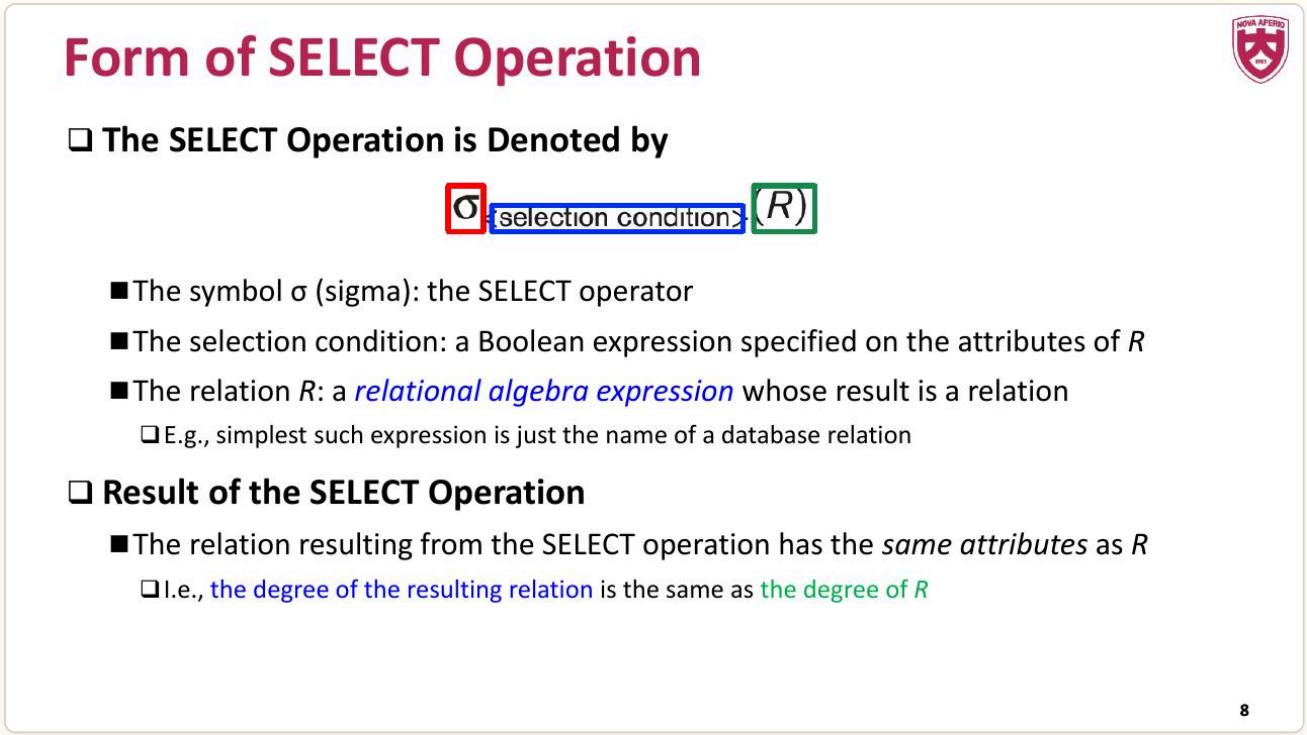
- **한 릴레이션**에서 선택 조건을 만족하는 튜플의 부분집합을 고른다
  - 조건을 만족하는 튜플만 남기고 나머지는 버리는 **필터**
- **가로 분할(horizontal partition)**로 그릴 수 있음
  - 조건을 만족(선택됨) vs 만족 안 함(걸러짐)

```
σ Dno=4 (EMPLOYEE)
σ Salary>30000 (EMPLOYEE)
```

✍ 메모

★ 시험 σ = 행(가로) 고르기, π = 열(세로) 고르기

⚠ 주의 관계 대수 SELECT(σ) ≠ SQL의 SELECT 절 (SQL SELECT 절은 π에 해당!)



SELECT의 형태

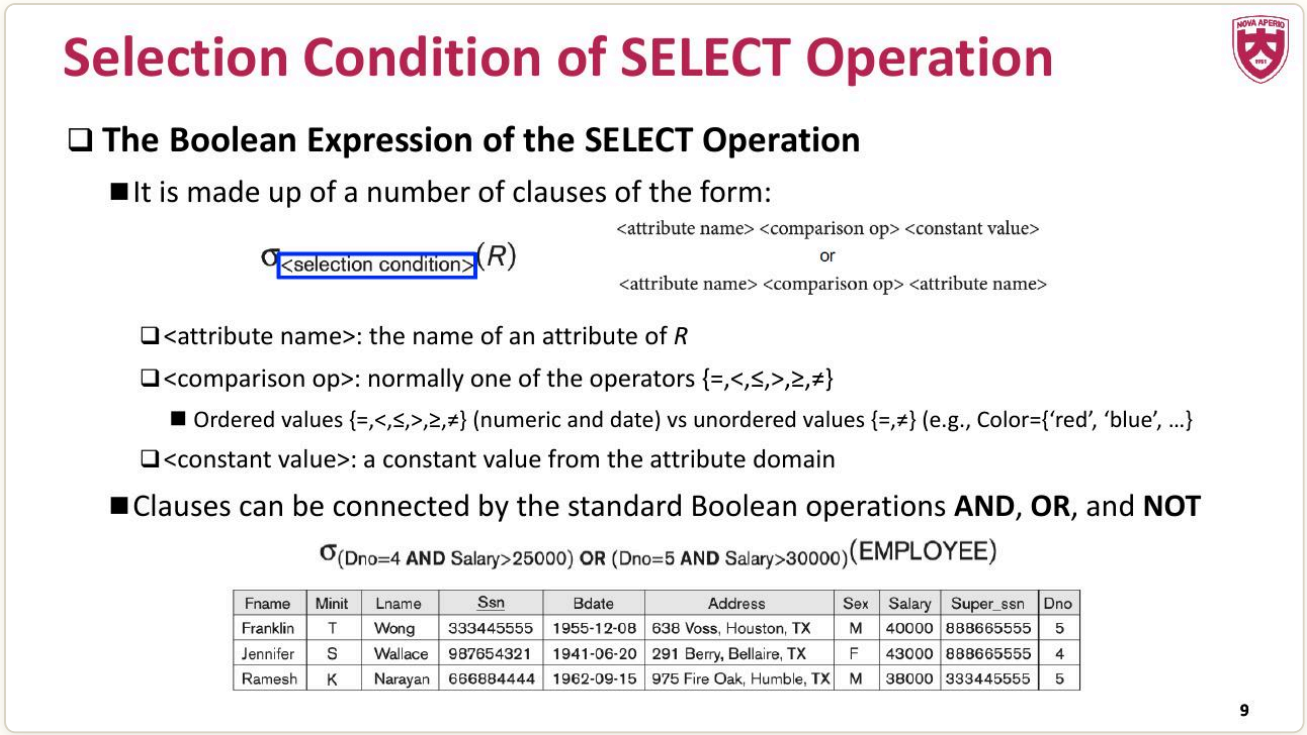
```
σ <selection condition> (R)
```

- **σ(sigma)**: SELECT 연산자
- 선택 조건: R의 속성에 대한 Boolean 식
- R: 결과가 릴레이션인 관계 대수식 (가장 단순하게는 릴레이션 이름)

- 결과
  - R과 **같은 속성** — 결과의 **차수(degree) = R의 차수**

✍ 메모

★ 시험 σ 결과의 degree = 원래와 같음, 튜플 수 ≤ 원래



Selection Condition of SELECT Operation

❑ The Boolean Expression of the SELECT Operation

■ It is made up of a number of clauses of the form:

$\sigma_{\text{<selection condition>}}(R)$

<attribute name> <comparison op> <constant value>
or
<attribute name> <comparison op> <attribute name>

❑ <attribute name>: the name of an attribute of *R*

❑ <comparison op>: normally one of the operators {=,<,≤,>,≥,≠}

■ Ordered values {=,<,≤,>,≥,≠} (numeric and date) vs unordered values {=,≠} (e.g., Color={'red', 'blue', ...}

❑ <constant value>: a constant value from the attribute domain

■ Clauses can be connected by the standard Boolean operations **AND**, **OR**, and **NOT**

$\sigma_{(\text{Dno=4 AND Salary>25000}) \text{ OR } (\text{Dno=5 AND Salary>30000})}(\text{EMPLOYEE})$

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

9

SELECT의 선택 조건

- 조건 절의 형태

```
<attribute name> <comparison op> <constant value>
<attribute name> <comparison op> <attribute name>
```

  - 비교 연산자: {=, <, ≤, >, ≥, ≠}
    - 순서 있는 값(숫자·날짜)은 모두 / 순서 없는 값(Color = {'red', 'blue', ...})은 {=, ≠}만
  - 절들은 **AND, OR, NOT**으로 연결

```
σ (Dno=4 AND Salary>25000) OR (Dno=5 AND Salary>30000) (EMPLOYEE)
```

✍ 메모

— 넘어가도 되는 쪽

Properties of SELECT Operation

❑ **The SELECT Operator is Unary**

- Being applied to "a single relation"
- It is applied to *each tuple individually*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\sigma_{Dno=4}(EMPLOYEE)$

❑ **The SELECT Operation is *Commutative***

- A sequence of SELECTs can be applied in an order

$$\sigma_{<cond1>}(\sigma_{<cond2>}(R)) = \sigma_{<cond2>}(\sigma_{<cond1>}(R))$$

- A sequence of SELECTs can be combined into a single SELECT

$$\sigma_{<cond1>}(\sigma_{<cond2>}(\ldots(\sigma_{<condn>}(R))\ldots)) = \sigma_{<cond1>\ \mathbf{AND}\ <cond2>\ \mathbf{AND}\ldots\mathbf{AND}\ <condn>}(R)$$

❑ **Correspondence to SQL**

$\sigma_{Dno=4\ \mathbf{AND}\ Salary>25000}(EMPLOYEE)$ ⟺

```
SELECT  *  Containing all attributes of R
FROM    EMPLOYEE
WHERE   Dno=4 AND Salary>25000;
```

The SELECT condition is specified in the WHERE clause of a query

10

SELECT의 성질

- **단항(unary)** — **한 릴레이션**에 적용, 각 튜플에 **개별**로
- **교환 가능(commutative)**

```
σ <c1> (σ <c2> (R)) = σ <c2> (σ <c1> (R))
```

  - 여러 SELECT를 **하나로** 합칠 수 있다

```
σ <c1> (σ <c2> (… σ <cn> (R))) = σ <c1> AND <c2> AND … AND <cn> (R)
```

- SQL 대응

```
σ Dno=4 AND Salary>25000 (EMPLOYEE)  ↔  SELECT * FROM EMPLOYEE WHERE Dno = 4 AND Salary > 25000;
```

✍ 메모

★ 시험 선택 조건 ↔ SQL WHERE 절

Unary Relational Operations: PROJECT

❑ What is the PROJECT Operation?

- ■ To select certain columns from the table and discards the other columns
  - ❑ SELECT: To select certain rows from the table and discards the other rows

$$\pi_{<\text{attribute list}>}(R)$$

- ■ Can be visualized as a *vertical partition* into two relations
  - ❑ One having the needed columns vs the other having the discarded columns

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\pi_{\text{Lname, Fname, Salary}}(\text{EMPLOYEE})$

Lname	Fname	Salary
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

11

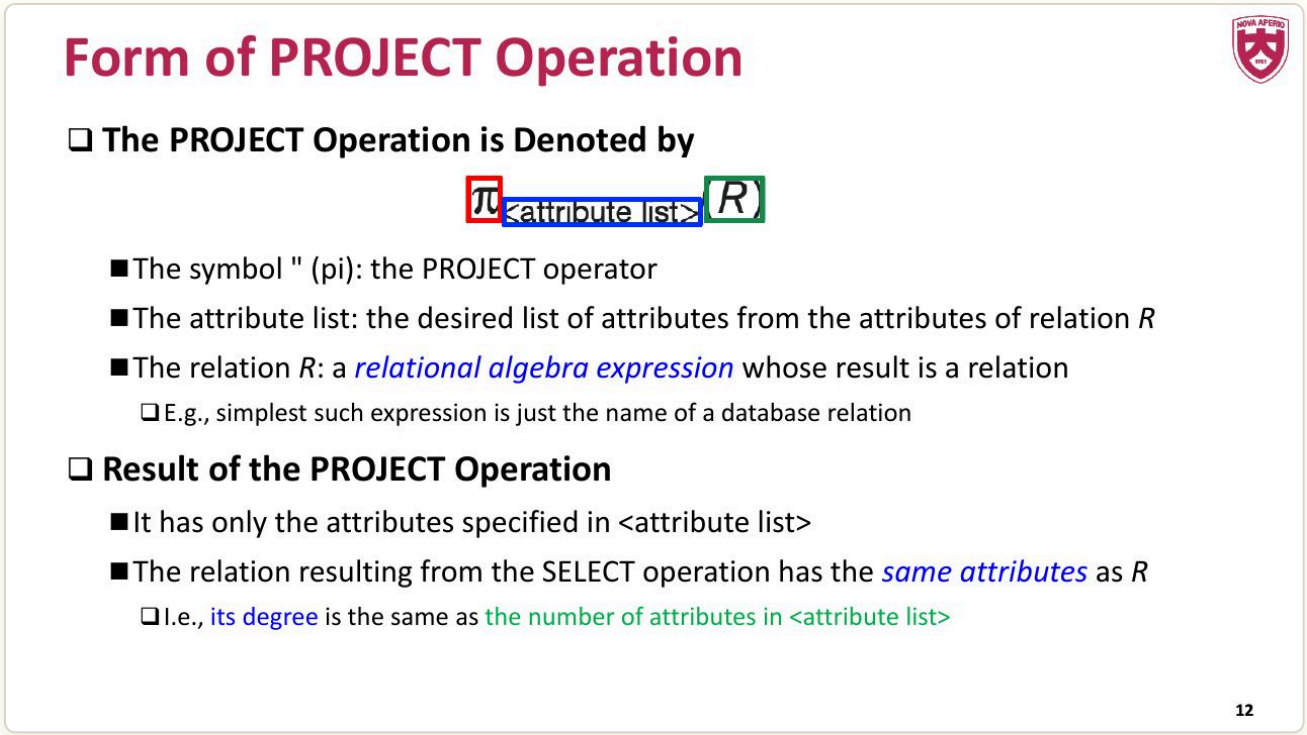
단항 연산: PROJECT (π)

- 테이블에서 **특정 열을 고르고** 나머지 열은 버린다
  - (SELECT는 특정 **행**을 고름)
- **세로 분할(vertical partition)**로 그릴 수 있음

```
π Lname, Fname, Salary (EMPLOYEE)
```

✍ 메모

— 넘어가도 되는 쪽



PROJECT의 형태

```
π <attribute list> (R)
```

- **π(pi)**: PROJECT 연산자
- 속성 목록: R의 속성 중 원하는 것

- 결과
  - <attribute list>의 속성만 가짐
  - **차수 = 속성 목록의 개수**

✍ 메모

💡 이해 슬라이드의 'resulting from the SELECT ... same attributes as R' 줄은 앞 쪽에서 남은 오타 — PROJECT는 속성 목록만

Properties of PROJECT Operation

❑ Duplicate Elimination

- Duplicate tuples are likely to occur *if the attribute list includes only nonkey attributes*
- The PROJECT operation removes any duplicate tuples
  - I.e., the result of the PROJECT operation is a "***set***" of distinct tuples

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
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Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\pi_{Sex,\ Salary}(EMPLOYEE)$

Sex	Salary
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

Set of distinct tuples

❑ Correspondence to SQL

$\pi_{Sex,\ Salary}(EMPLOYEE)$ ⟺

```
SELECT   DISTINCT Sex, Salary
FROM     EMPLOYEE
```

The PROJECT attribute list is specified in the SELECT clause of a query

13

PROJECT의 성질 — 중복 제거

- 키가 아닌 속성만 고르면 **중복 튜플**이 생기기 쉽다
  - PROJECT는 **중복 튜플을 제거**
  - 결과는 서로 다른 튜플의 **집합(set)**

```
π Sex, Salary (EMPLOYEE)  → (F, 25000)이 한 번만
```

- SQL 대응

```
π Sex, Salary (EMPLOYEE)  ↔  SELECT DISTINCT Sex, Salary FROM EMPLOYEE;
```

✍ 메모

★ 시험 π는 중복 제거 → SQL에선 DISTINCT를 붙여야 같다

⚠ 주의 결과 튜플 수 ≤ 원래 (중복이 사라지니까)

Sequences of Relational Operations

❑ What If Need to Apply Several Relational Algebra Operations?

Q. Retrieve the first name, last name, and salary of all employees who work in department number 5

One possible way:
Explicitly show the sequence of operations by **giving a name to each intermediate relation**

$$DEP5_EMPS \leftarrow \sigma_{Dno=5}(EMPLOYEE)$$
$$RESULT \leftarrow \pi_{Fname,\ Lname,\ Salary}(DEP5_EMPS)$$

Using the assignment operation: ← (left arrow)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

⇒

DEP5_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston,TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston,TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

⇒

RESULT

Fname	Lname	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

It is usually simpler to break down a complex sequence of operations by specifying intermediate result relations than to write a single relational algebra expression

14

연산 잇기 ① 하나의 식

- 여러 연산을 써야 하면?
- Q. 5번 부서 직원의 이름, 성, 월급

```
π Fname, Lname, Salary (σ Dno=5 (EMPLOYEE))
```

  - 식 안에 식을 넣어 **한 줄로**

✍ 메모

— 넘어가도 되는 쪽

Sequences of Relational Operations (cont.)

❑ What If Need to Apply Several Relational Algebra Operations?

Q. Retrieve the first name, last name, and salary of all employees who work in department number 5

Another way:
Write the operations as a **single relational algebra expression** by nesting the operations

$$\pi_{Fname,\ Lname,\ Salary}(\sigma_{Dno=5}(EMPLOYEE))$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Fname	Lname	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

15

연산 잇기 ② 중간 결과에 이름

- **대입 연산(←)**으로 중간 결과에 이름을 붙인다

```
DEP5_EMPS ← σ Dno=5 (EMPLOYEE)
RESULT ← π Fname, Lname, Salary (DEP5_EMPS)
```

- 복잡한 연산은 **중간 릴레이션을 명시**하는 편이 한 식보다 보통 더 쉽다

✍ 메모

💡 이해 시험 답안은 이 방식이 실수가 적다 (정답 슬라이드들도 이 형식)

RENAME Operation

❑ Defining a RENAME Operation As a Unary Operator

- ■ To rename either the relation name or the attribute names, or both
- ■ The RENAME operation is denoted by any of the following three forms:

$$\rho_{S(B1, B2, \ldots, Bn)}(R) \quad \text{or} \quad \rho_{S}(R) \quad \text{or} \quad \rho_{(B1, B2, \ldots, Bn)}(R)$$

Both the relation and its attributes | The relation only | The attributes only

- ❑ The symbol ρ (rho): the RENAME operator
- ❑ S: the new relation name
- ❑ B_i : the new attribute name ($1 \le i \le n$, n is the degree of R)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

TEMP

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston,TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston,TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

$$\text{TEMP} \leftarrow \sigma_{Dno=5}(\text{EMPLOYEE})$$

$$R(\text{First_name, Last_name, Salary}) \leftarrow \pi_{\text{Fname, Lname, Salary}}(\text{TEMP})$$

We can also rename the attributes in the intermediate relations

16

RENAME (ρ)

- 릴레이션 이름이나 속성 이름, 또는 둘 다 바꾸는 **단항** 연산

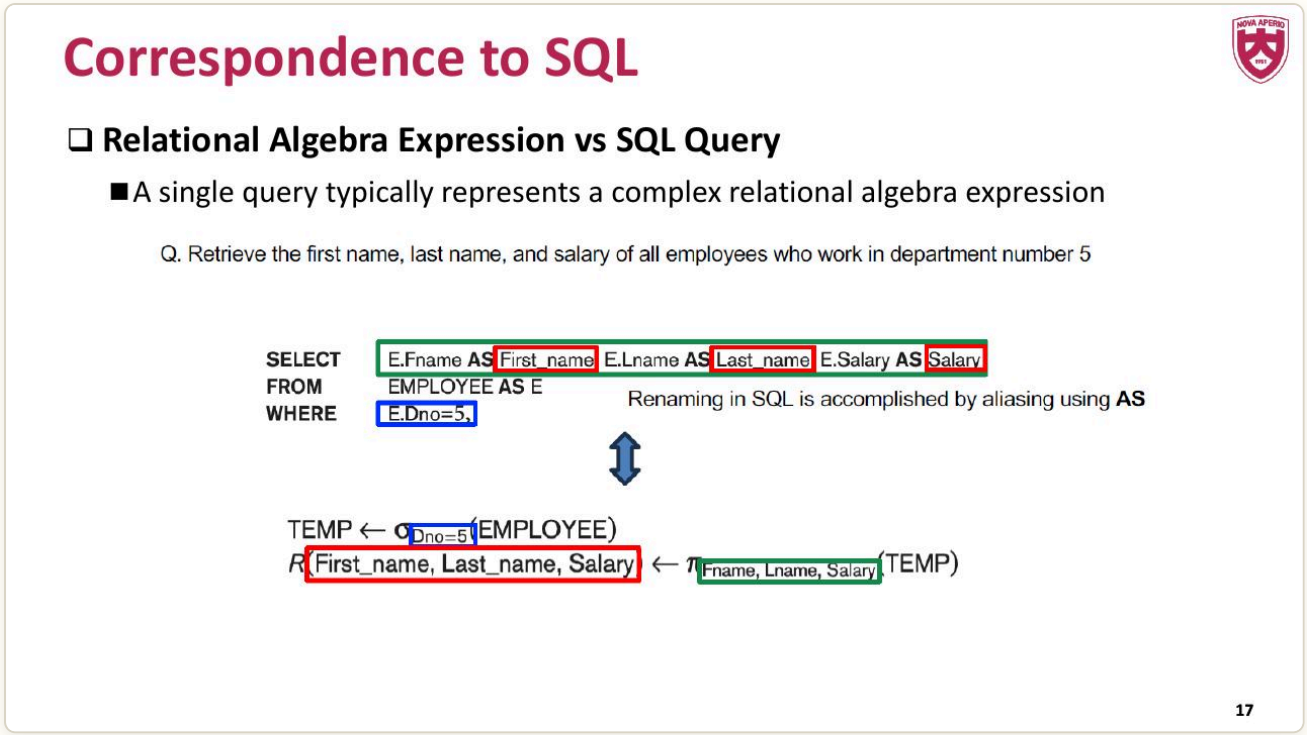
```
ρ S(B1, …, Bn) (R)   – 둘 다
ρ S (R)              – 릴레이션 이름만
ρ (B1, …, Bn) (R)    – 속성 이름만
```

  - **ρ(rho)**: RENAME 연산자 / S: 새 릴레이션 이름 / B_i: 새 속성 이름 (n = R의 차수)
- 중간 결과에서도 이름 바꾸기 가능

```
TEMP ← σ Dno=5 (EMPLOYEE)
R(First_name, Last_name, Salary) ← π Fname, Lname, Salary (TEMP)
```

✍ 메모

★ 시험 ρ 표기 3가지



SQL과의 대응

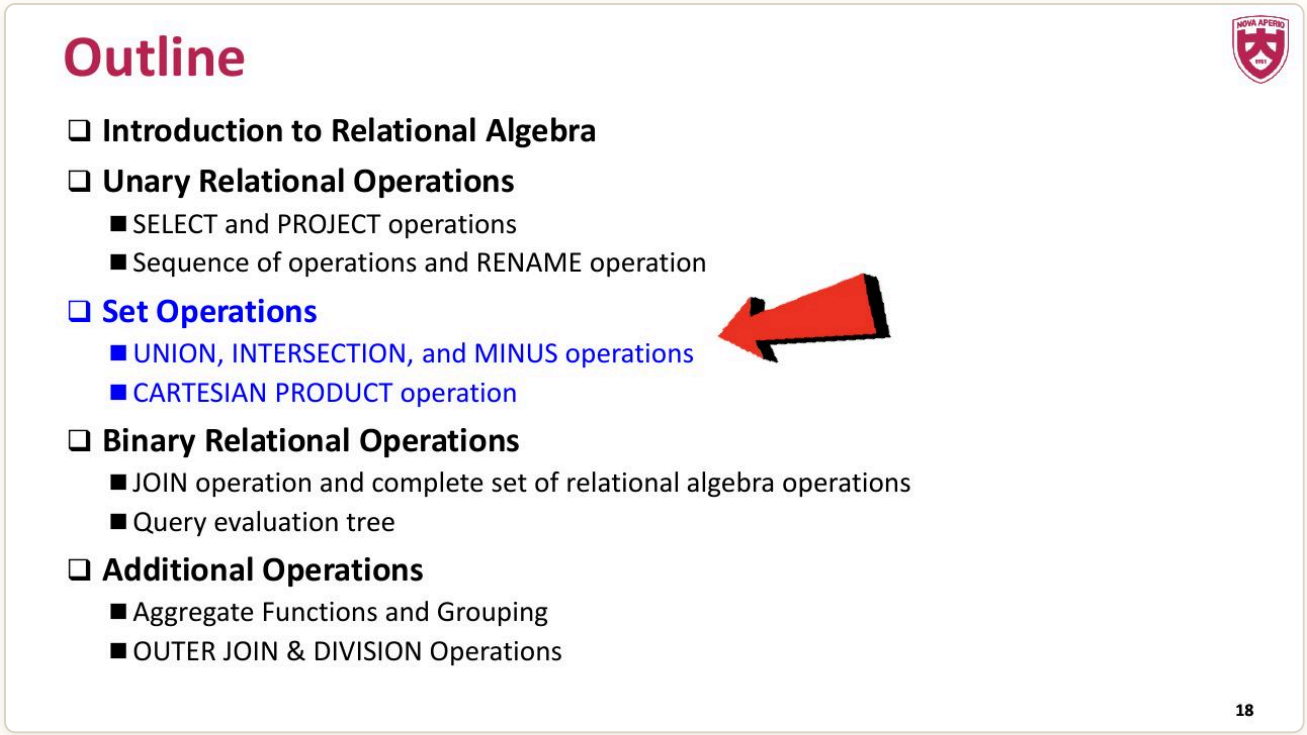
- 관계 대수식 vs SQL 질의

```
SELECT E.Fname AS First_name, E.Lname AS Last_name, E.Salary AS Salary FROM EMPLOYEE AS E
WHERE E.Dno = 5;
```

  - SQL의 rename = **AS 별칭**
  - SQL 질의 하나는 보통 **복잡한 관계 대수식** 하나에 해당

✍ 메모

— 넘어가도 되는 쪽



목차 (Outline)

- **관계 대수 소개**
- **단항 연산** — SELECT·PROJECT / 연산 잇기와 RENAME
- **집합 연산** — UNION·INTERSECTION·MINUS / CARTESIAN PRODUCT
- **이항 연산** — JOIN과 완전 집합 / 질의 평가 트리
- **추가 연산** — 집계 함수와 그룹화 / OUTER JOIN과 DIVISION

✍️ 메모

💡 이해 집합 연산

Set Operations

❑ Set Operations as Relational Algebra Operations

Q. Retrieve the SSNs of all employees who either work in department 5 or directly supervise an employee who works in department 5

DEP5_EMPS ← $\sigma_{Dno=5}$(EMPLOYEE)
RESULT1 ← π_{Ssn}(DEP5_EMPS)
RESULT2(Ssn) ← π_{Super_ssn}(DEP5_EMPS)
RESULT ← RESULT1 ∪ RESULT2

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

RESULT1

Ssn
123456789
333445555
666884444
453453453

RESULT2

Ssn
333445555
888665555

RESULT

Ssn
123456789
333445555
666884444
453453453
888665555

The SSN value '33344555' appears only once

❑ Types of Set Operations

■ UNION, INTERSECTION, SET DIFFERENCE (MINUS or EXCEPT)

❑ Binary operations, each of which is applied to two sets

19

집합 연산

- 관계 대수의 집합 연산
  - **UNION, INTERSECTION, SET DIFFERENCE** (MINUS 또는 EXCEPT)
  - 두 집합에 적용하는 **이항** 연산

✍ 메모

— 넘어가도 되는 쪽

Set Operations (Cont.)

❑ Type Compatibility (or Union Compatibility)

■ Two relations $R(A_1,..., A_n)$ and $S(B_1,..., B_n)$ are said to be *type compatible*:

If $\text{dom}(A_i) = \text{dom}(B_i)$ for $1 \le i \le n$

The two relations have the same number of attributes,
and each corresponding pair of attributes has the same domain

❑ UNION Operation

■ The result is a relation including all tuples that are either **in *R* or in *S* or in both**

❑ Denoted by $R \cup S$, *duplicate tuples are eliminated*

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT ∪ INSTRUCTOR

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

STUDENT and INSTRUCTOR are type-compatible

The names of all students and instructors

20

Type Compatibility와 UNION

- **Type(Union) Compatibility**: R(A1, ..., An)과 S(B1, ..., Bn)이
  - ① 속성 수가 같고 ② $\text{dom}(A_i) = \text{dom}(B_i)$ $(1 \le i \le n)$
- **UNION R ∪ S**: R에 있거나 S에 있거나 둘 다에 있는 모든 튜플
  - **중복 제거**
- 예: STUDENT ∪ INSTRUCTOR = 모든 학생과 강사의 이름

✍ 메모

★ 시험 type compatible 조건 2개 (속성 수, 대응 도메인)

Set Operations (Cont.)

❑ INTERSECTION Operation

■ The result is a relation including all tuples that are in both *R* and *S*

❑ Denoted by $R \cap S$

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT ∩ INSTRUCTOR

Fn	Ln
Susan	Yao
Ramesh	Shah

Only those who are both students and instructors

STUDENT and INSTRUCTOR are type-compatible

❑ Properties of UNION and INTERSECTION

■ $R \cup S = S \cup R$, $R \cap S = S \cap R$ (i.e., *Commutative*)

■ $R \cup (S \cup T) = (S \cup R) \cup T$, $R \cap (S \cap T) = (S \cap R) \cap T$ (i.e., *Associative*)

21

INTERSECTION과 성질

- **INTERSECTION R ∩ S**: R과 S **둘 다에** 있는 튜플
- ∪·∩의 성질
  - **교환**: R ∪ S = S ∪ R, R ∩ S = S ∩ R
  - **결합**: R ∪ (S ∪ T) = (R ∪ S) ∪ T, R ∩ (S ∩ T) = (R ∩ S) ∩ T

✍ 메모

— 넘어가도 되는 쪽

Set Operations (Cont.)

❑ SET DIFFERENCE (MINUS) Operation

■ The result is a relation including all tuples that are in *R* but not in *S*

❑ Denoted by $R - S$

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT – INSTRUCTOR

Fn	Ln
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR – STUDENT

Fname	Lname
John	Smith
Ricardo	Browne
Francis	Johnson

❑ The MINUS Operation is Not *Commutative*

■ $R - S \neq S - R$

22

SET DIFFERENCE (MINUS)

- **R – S**: R에는 있고 S에는 **없는** 튜플
- MINUS는 **교환 안 됨**: R – S ≠ S – R
- 예: STUDENT – INSTRUCTOR = {Johnny Kohler, Barbara Jones, Amy Ford, Jimmy Wang, Ernest Gilbert}
  - INSTRUCTOR – STUDENT = {John Smith, Ricardo Browne, Francis Johnson}

✍ 메모

★ 시험 ∪ ∩ 는 교환 가능, – 는 불가

CARTESIAN PRODUCT

❑ What is the CARTESIAN PRODUCT Operation?

- ■ A Binary set operation, which can be applied to not type-compatible relations
  - ❑ a.k.a **CROSS PRODUCT** or **CROSS JOIN**
- ■ Produce new tuples by concatenating *all possible combinations of tuples*
  - ❑ Denoted by $R(A_1,..., A_n)\ X\ S(B_1,..., B_m)$
  - ❑ The result: $Q(A_1,..., A_n, B_1,..., B_m)$
    - ■ The number of attributes: n + m
    - ■ The number of tuples: |R| * |S| (all possible combinations)

❑ Usage of the CARTESIAN PRODUCT

- ■ This operation alone is generally *meaningless*
- ■ Useful when followed by a *selection matching certain values of attributes*

23

CARTESIAN PRODUCT (×)

- **type-compatible이 아니어도** 적용 가능한 이항 집합 연산 — a.k.a CROSS PRODUCT, CROSS JOIN
- 가능한 **모든 튜플 조합**을 이어 붙인다

```
R(A1, …, An) × S(B1, …, Bm)  →  Q(A1, …, An, B1, …, Bm)
```

  - 속성 수: **n + m** / 튜플 수: **|R| × |S|**
- 혼자서는 대개 **무의미** — 뒤에 값을 맞추는 **선택**이 따를 때 유용

✍ 메모

★ 시험 속성 n+m, 튜플 |R|×|S| — 계산 문제 단골

CARTESIAN PRODUCT: Example

❑ When the CARTESIAN PRODUCT Useful?

Q. Retrieve a list of names of each female employee's dependents

FEMALE_EMPS ← $\sigma_{Sex='F'}$(EMPLOYEE)
EMPNAMES ← $\pi_{Fname, Lname, Ssn}$(FEMALE_EMPS)
EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
ACTUAL_DEPENDENTS ← $\sigma_{Ssn=Essn}$(EMP_DEPENDENTS)
RESULT ← $\pi_{Fname, Lname, Dependent_name}$(ACTUAL_DEPENDENTS)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

FEMALE_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	1968-07-19	3321Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291Berry, Bellaire, TX	F	43000	888665555	4
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

24

카티션 곱 예 ①

- Q. 여직원의 부양가족 이름

```
FEMALE_EMPS ← σ Sex='F' (EMPLOYEE)
EMPNAMES ← π Fname, Lname, Ssn (FEMALE_EMPS)
```

  - → Alicia Zelaya, Jennifer Wallace, Joyce English (3개)

✍ 메모

— 넘어가도 되는 쪽

CARTESIAN PRODUCT: Example (cont.)

❑ When the CARTESIAN PRODUCT Useful?

Q. Retrieve a list of names of each female employee's dependents

FEMALE_EMPS ← $\sigma_{Sex='F'}$(EMPLOYEE)
EMPNAMES ← $\pi_{Fname, Lname, Ssn}$(FEMALE_EMPS)
EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
ACTUAL_DEPENDENTS ← $\sigma_{Ssn=Essn}$(EMP_DEPENDENTS)
RESULT ← $\pi_{Fname, Lname, Dependent_name}$(ACTUAL_DEPENDENTS)

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

3 tuples

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

7 tuples

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

3*7 tuples

Every tuple from EMPNAMES is combined with every tuple from DEPENDENT
This result is not that meaningful!

25

카티션 곱 예 ②

```
EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
```

- 3 × 7 = **21개** 튜플 — 대부분 서로 관계없는 조합 (의미 없음)

✍ 메모

✏ 예·풀이 직원 3명 × 부양가족 7명 = 21행

CARTESIAN PRODUCT: Example (cont.)

❑ When the CARTESIAN PRODUCT Useful?

Q. Retrieve a list of names of each female employee's dependents

FEMALE_EMPS ← $\sigma_{Sex='F'}$(EMPLOYEE)
EMPNAMES ← $\pi_{Fname, Lname, Ssn}$(FEMALE_EMPS)
EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
ACTUAL_DEPENDENTS ← $\sigma_{Ssn=Essn}$(EMP_DEPENDENTS)
RESULT ← $\pi_{Fname, Lname, Dependent_name}$(ACTUAL_DEPENDENTS)

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

We want to combine a female employee tuple only with her particular dependents

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

RESULT

Fname	Lname	Dependent_name
Jennifer	Wallace	Abner

26

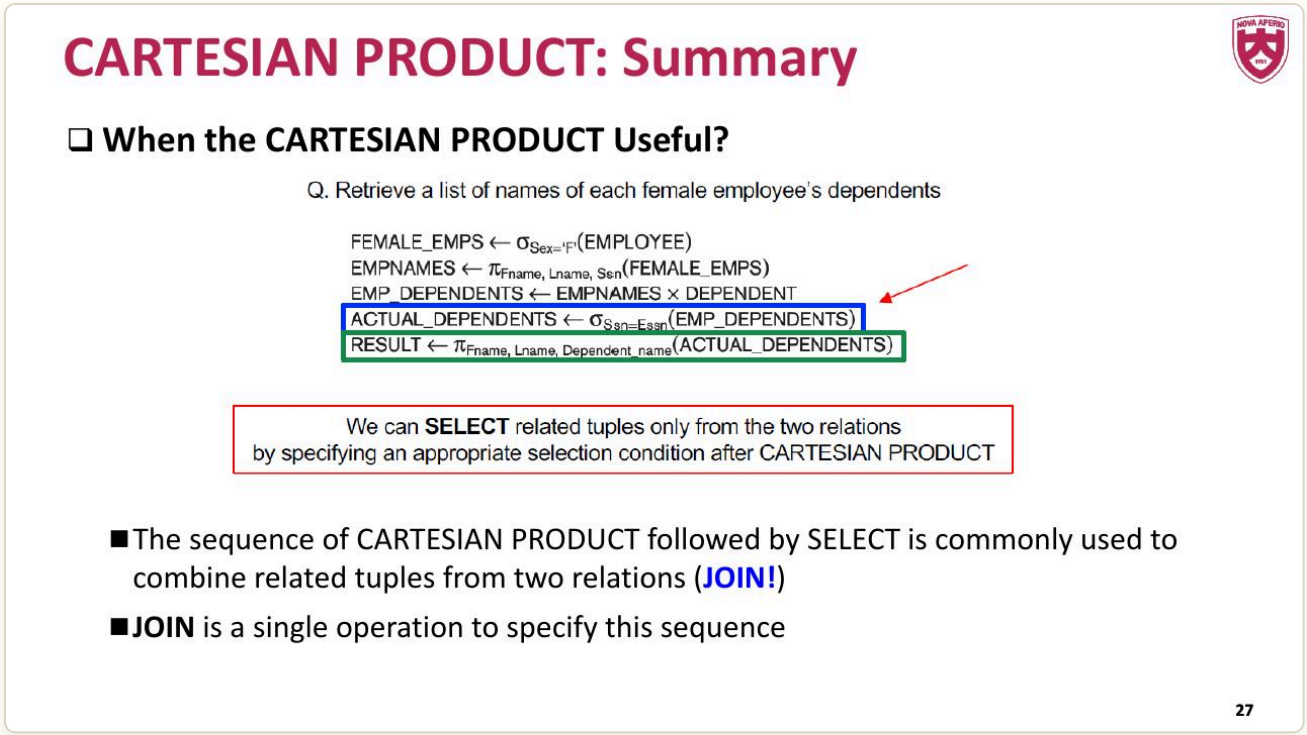
카티션 곱 예 ③

```
ACTUAL_DEPENDENTS ← σ Ssn=Essn (EMP_DEPENDENTS)
RESULT ← π Fname, Lname, Dependent_name (ACTUAL_DEPENDENTS)
```

- → **Jennifer Wallace, Abner**

✍ 메모

★ 시험 × 다음 σ(같은 값 맞추기) = 진짜 짝만 남음 → 이게 JOIN

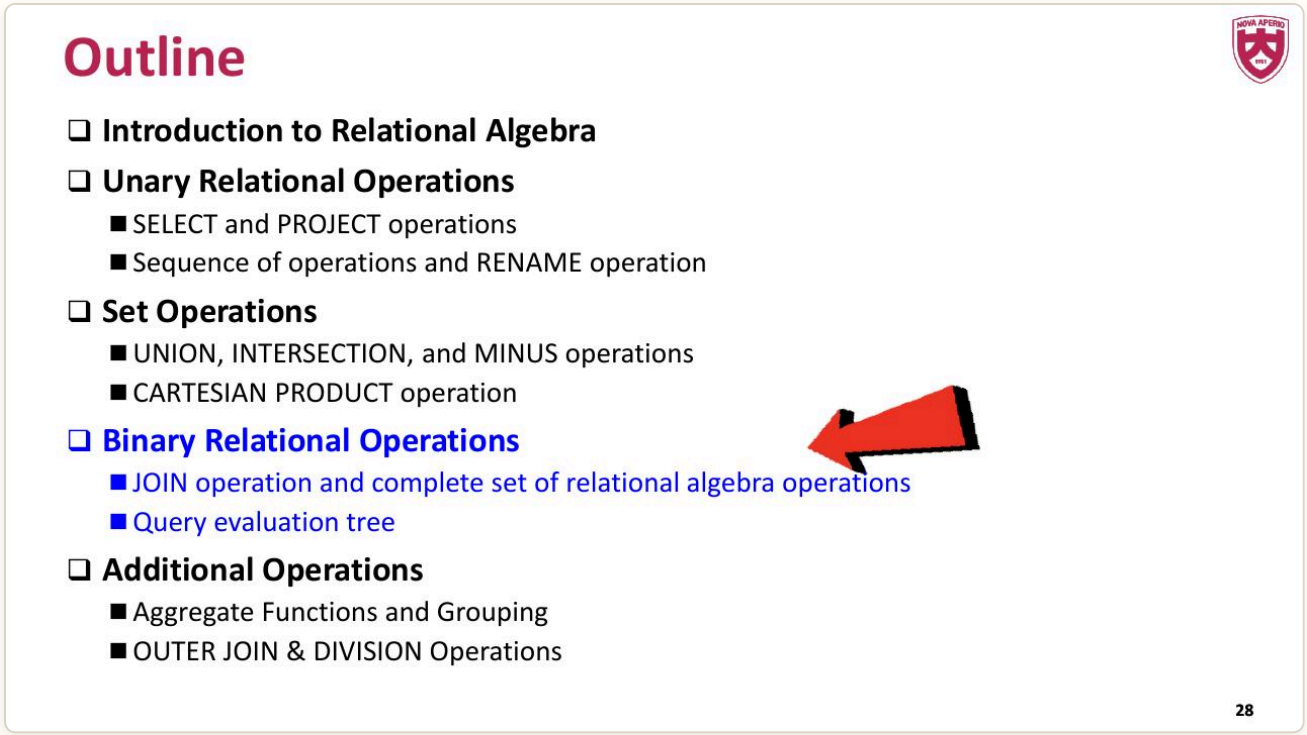


카티션 곱 정리

- **카티션 곱 다음 SELECT**는 두 릴레이션의 관련 튜플을 결합하는 흔한 순서 (**JOIN!**)
  - JOIN = 이 순서를 **한 연산**으로

✍ 메모

— 넘어가도 되는 쪽



목차 (Outline)

- **관계 대수 소개**
- **단항 연산** — SELECT·PROJECT / 연산 잇기와 RENAME
- **집합 연산** — UNION·INTERSECTION·MINUS / CARTESIAN PRODUCT
- **이항 연산** — JOIN과 완전 집합 / 질의 평가 트리
- **추가 연산** — 집계 함수와 그룹화 / OUTER JOIN과 DIVISION

✍ 메모

💡 이해 이항 연산: JOIN

Binary Relational Operations: JOIN

❑ What is the JOIN Operation?

■ Combine *related tuples* from two relations into single "longer" tuples

$$R \bowtie_{\text{<join condition>}} S$$

Q. Retrieve the name of the manager of each department

DEPT_MGR ← DEPARTMENT $\bowtie_{Mgr_ssn=Ssn}$ EMPLOYEE

RESULT ← $\pi_{Dname, Lname, Fname}$(DEPT_MGR)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Mgr_ssn is a foreign key of the DEPARTMENT that references Ssn, the primary key of the EMPLOYEE

DEPT_MGR

Dname	Dnumber	Mgr_ssn	...	Fname	Minit	Lname	Ssn	...
Research	5	333445555	...	Franklin	T	Wong	333445555	...
Administration	4	987654321	...	Jennifer	S	Wallace	987654321	...
Headquarters	1	888665555	...	James	E	Borg	888665555	...

Combine each department tuple with the employee tuple whose Ssn value matches the Mgr_ssn value in the department tuple

29

이항 연산: JOIN (⋈)

- 두 릴레이션의 **관련 튜플을 하나의 더 긴 튜플로** 결합

```
DEPT_MGR ← DEPARTMENT ⋈ Mgr_ssn=Ssn EMPLOYEE
```

✍ 메모

— 넘어가도 되는 쪽

Binary Relational Operations: JOIN (cont.)



❑ Comparison of JOIN with CARTESIAN PRODUCT

■ The JOIN can be specified as a CARTESIAN PRODUCT followed by SELECT

EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
ACTUAL_DEPENDENTS ← $\sigma_{Ssn=Essn}$(EMP_DEPENDENTS)



ACTUAL_DEPENDENTS ← EMPNAMES $\bowtie_{Ssn=Essn}$ DEPENDENT

These two operations can be replaced with a single JOIN operation

30

JOIN과 카티션 곱 비교

- JOIN = **카티션 곱 후 SELECT**

```
EMP_DEPENDENTS ← EMPNAMES × DEPENDENT
ACTUAL_DEPENDENTS ← σ Ssn=Essn (EMP_DEPENDENTS)
⟹ ACTUAL_DEPENDENTS ← EMPNAMES ⋈ Ssn=Essn DEPENDENT
```

✍ 메모

★ 시험 $R \bowtie_C S \equiv \sigma_C(R \times S)$

Form of JOIN Operation



❑ The JOIN Operation is Denoted by

$R \bowtie_{\text{<join condition>}} S$

- ■ The symbol ⋈ (bowtie): the JOIN operator
- ■ The two relations: $R(A_1,..., A_n)$ and $S(B_1,..., B_m)$
- ■ The join condition: a Boolean expression specified on the attributes of R

❑ Result of the JOIN Operation: $Q(A_1,..., A_n, B_1,..., B_m)$

- ■ Contain only *the combinations of tuples satisfying the join condition*
  - ❑ The join condition is evaluated for each tuple combination;
  - ❑ The tuple combination is included in the result if it satisfies the condition
- ■ Whereas all combinations of tuples are included in the CARTESIAN PRODUCT

31

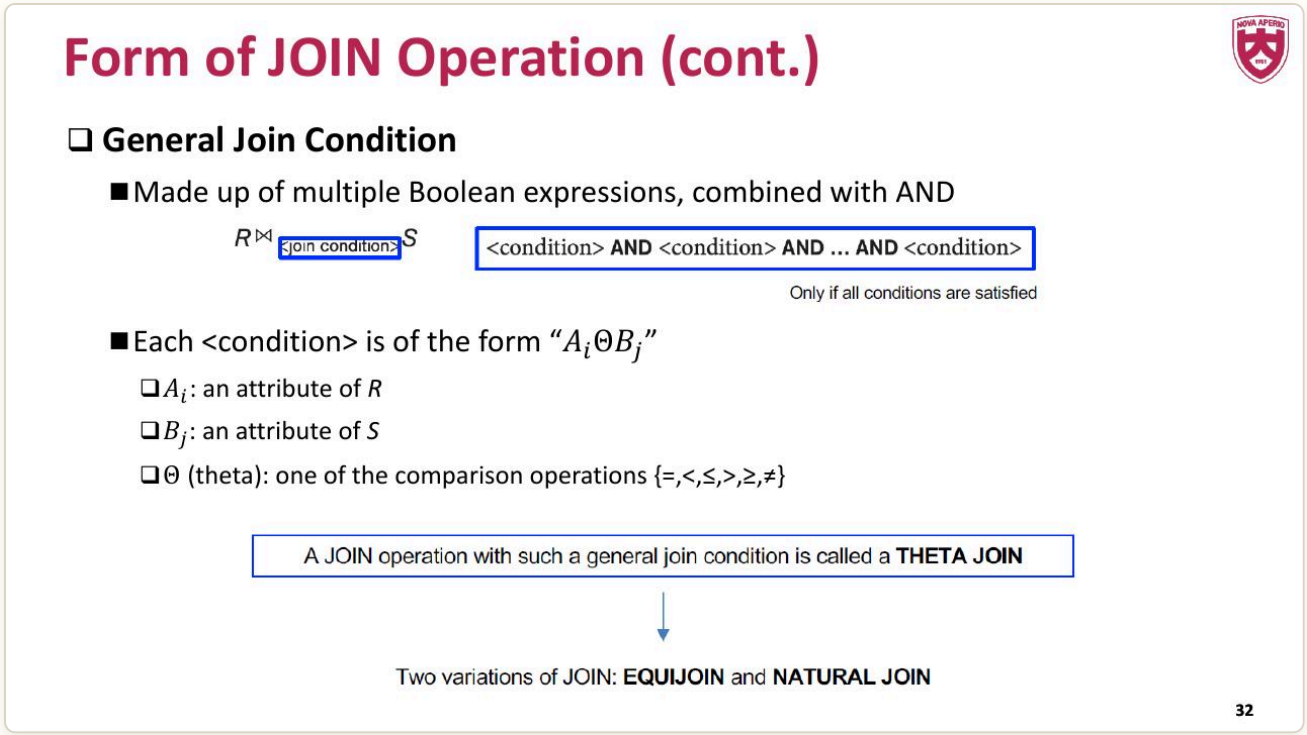
JOIN의 형태

```
R ⋈ <join condition> S
```

- ⋈(bowtie): JOIN 연산자 / R(A1, ..., An), S(B1, ..., Bm)
- 결과 Q(A1, ..., An, B1, ..., Bm)
  - 조인 조건을 **만족하는 조합만** 포함
    - 조합마다 조건을 평가, 만족하면 결과에
  - 카티션 곱은 **모든** 조합을 포함

✍ 메모

— 넘어가도 되는 쪽

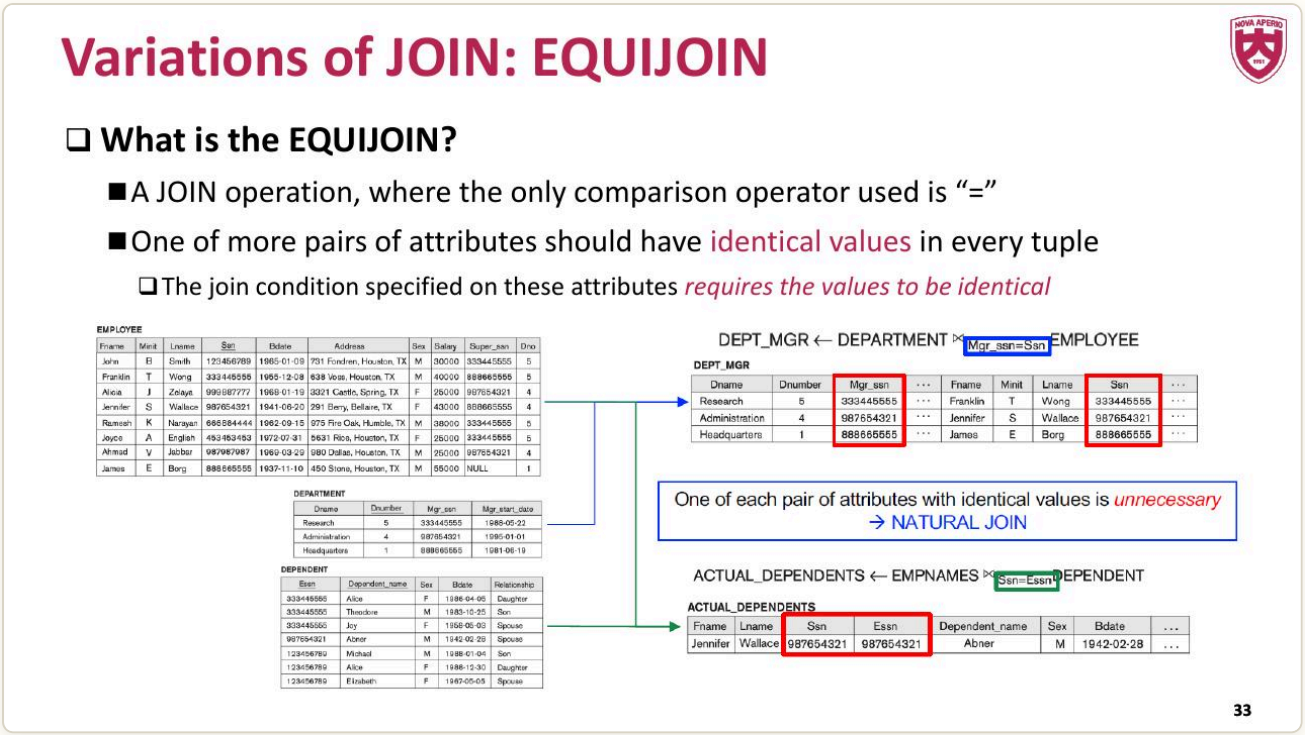


일반 조인 조건 — THETA JOIN

- 여러 Boolean 식을 **AND**로 연결
  - 각 조건은 **A_i θ B_j** 꼴
    - A_i: R의 속성 / B_j: S의 속성
    - **θ(theta)**: {=, <, ≤, >, ≥, ≠} 중 하나
  - → 이런 조인을 **theta join**이라 함

✍ 메모

— 넘어가도 되는 쪽

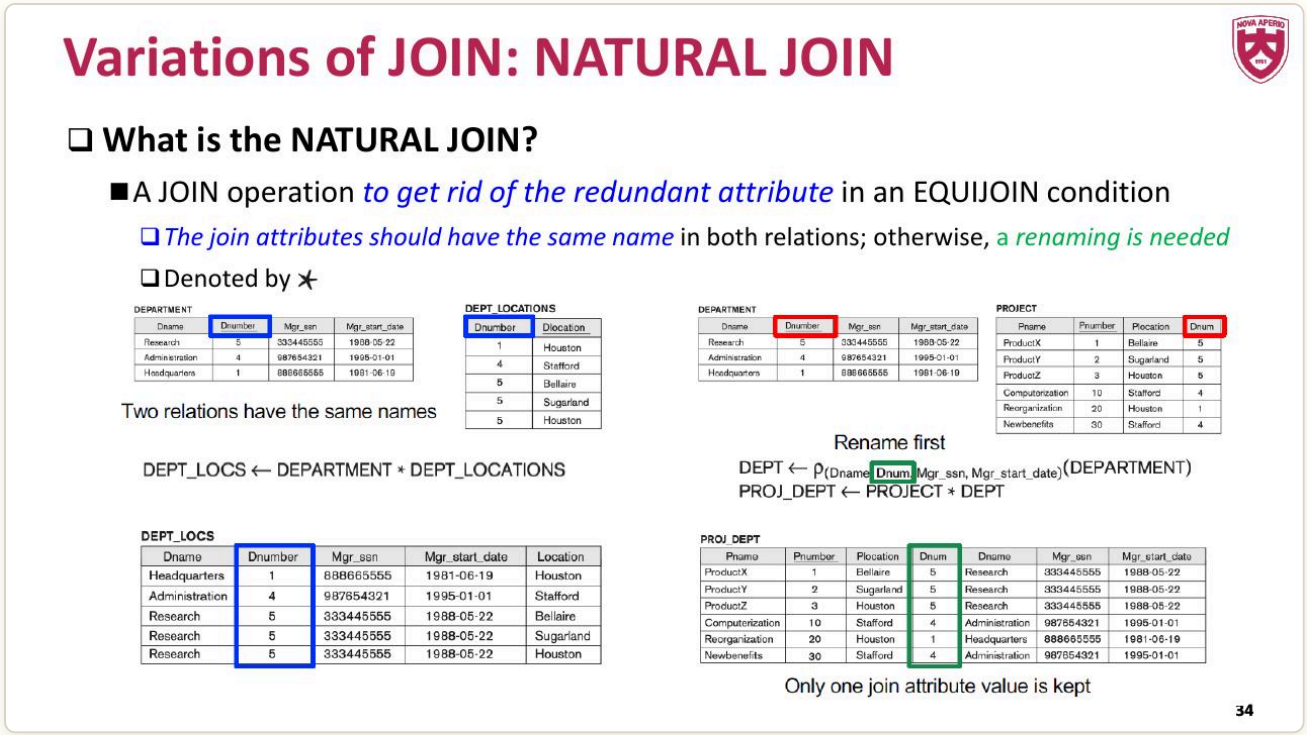


JOIN의 변형: EQUIJOIN

- 비교 연산자로 **'='만** 쓰는 JOIN
  - 한 쌍 이상의 속성이 모든 튜플에서 **값이 같다**
  - → 결과에 같은 값 속성 쌍이 **중복** (예: Mgr_ssn과 Ssn)

✍ 메모

💡 이해 같은 값 두 칸 중 하나는 쓸모없음 → 다음 쪽 NATURAL JOIN



JOIN의 변형: NATURAL JOIN (*)

- EQUIJOIN에서 **중복 속성을 없앤** JOIN
  - 조인 속성이 두 릴레이션에서 **같은 이름**이어야 — 아니면 **먼저 rename**

```
R * S
DEPT_LOCS ← DEPARTMENT * DEPT_LOCATIONS          (Dnumber 공유)
PROJ_DEPT ← PROJECT * ρ(Dname, Dnum, Mgr_ssn, Mgr_start_date) (DEPARTMENT)
```

✍ 메모

★ 시험 NATURAL JOIN = 같은 이름 속성으로 equijoin + 그 속성 하나만 남김

⚠ 주의 이름이 다르면 ρ로 먼저 맞춰야 (Dnumber → Dnum)

Properties of JOIN Operation



❑ The Size of the Result of JOIN Operation

$$Q \leftarrow R \bowtie_{\text{<join condition>}} S \qquad 0 \leq |Q| \leq |R| * |S|$$

- ■ (Minimum) If no combination of tuples satisfies the join condition
  - → The result is an empty relation
- ■ (Maximum) If all combinations of tuples satisfy the condition or no join condition
  - → The result includes all possible combinations of tuples (i.e., degeneralizing to a **CARTESIAN PRODUCT**)

❑ Join Selectivity of a Join Condition

$$\frac{\text{Expected Size of the Join Result}}{|R| * |S|}$$

35

JOIN의 성질 — 결과 크기

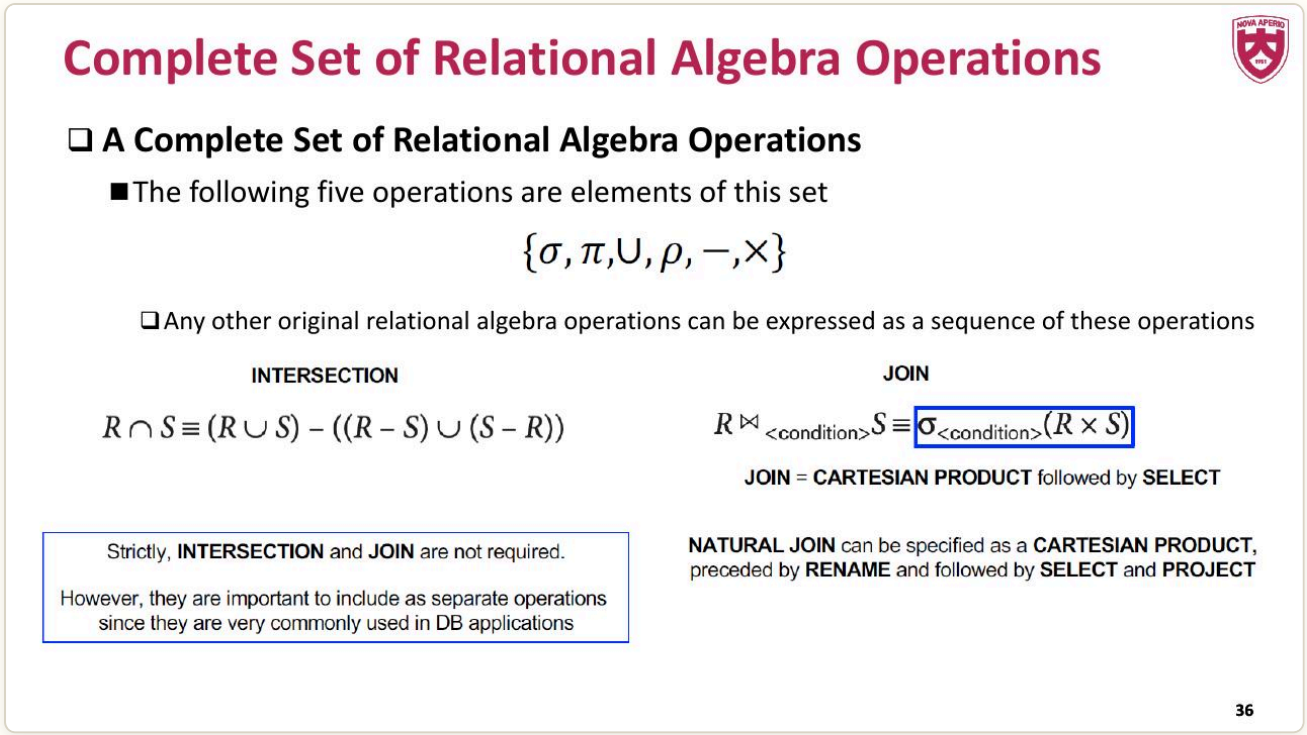
- (최소) 조건을 만족하는 조합이 없으면 → **빈 릴레이션**
- (최대) 모든 조합이 만족하거나 조건이 없으면 → **모든 조합** = 카티션 곱으로 퇴화

```
0 ≤ |R ⋈ S| ≤ |R| · |S|
```

- **Join selectivity** = 예상 조인 결과 크기 / (|R| · |S|)

✍ 메모

★ 시험 join selectivity 식 — 비율(0~1)



관계 대수의 완전 집합

- 다음 연산들이 **완전 집합**: **{σ, π, ∪, ρ, −, ×}**
  - 다른 모든 원래 관계 대수 연산은 이것들의 순서로 표현 가능

```
R ∩ S ≡ (R ∪ S) − ((R − S) ∪ (S − R))
R ⋈ <condition> S ≡ σ <condition> (R × S)
```

  - NATURAL JOIN = RENAME → CARTESIAN PRODUCT → SELECT → PROJECT
- 엄밀히는 ∩, JOIN이 필요 없지만 **자주 써서** 따로 둔다

✍ 메모

★ 시험 완전 집합 6개 기호 외우기 (슬라이드 문장은 'five operations'인데 기호는 여섯)

⚠ 주의 ∩와 ⋈는 완전 집합에 없다 — O/X 함정

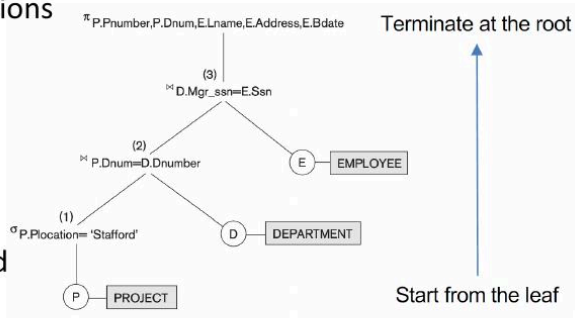
Query Evaluation Tree

❑ What is a Query Evaluation (Execution) Tree?

- ■ Tree data structure for the *internal representation of the query* in an RDMBS
  - ❑ Corresponding to a relational algebra expression (the relational algebra operations being executed)
- ■ Leaf nodes: the input relations
- ■ Internal nodes: the relational algebra operations

❑ Execution of the Query Tree

- ■ Executing an internal node operation
  - ❑ Whenever its operands (child nodes) are available
- ■ Replacing that internal node by the result
- ■ Repeat this process until the root is executed



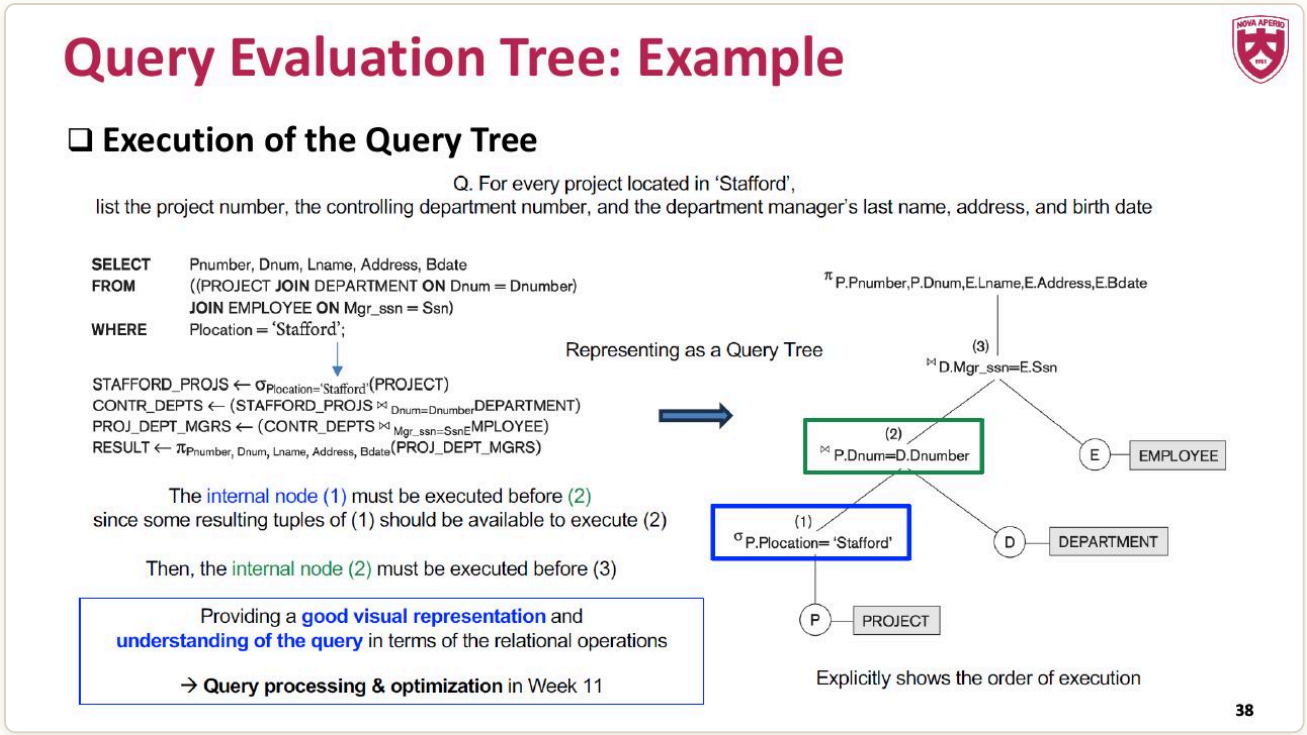
37

질의 평가 트리 (Query Evaluation Tree)

- RDBMS 안에서 질의를 내부 표현하는 **트리**
  - 관계 대수식에 대응 (실행되는 관계 대수 연산)
  - **잎 노드**: 입력 릴레이션 / **내부 노드**: 관계 대수 연산
- 실행
  - 피연산자(자식)가 준비되면 내부 노드 연산 실행
  - 그 노드를 결과로 바꿈
  - **루트까지** 반복

✍ 메모

★ 시험 잎 = 입력 표, 안쪽 = 연산, 아래에서 위로 실행

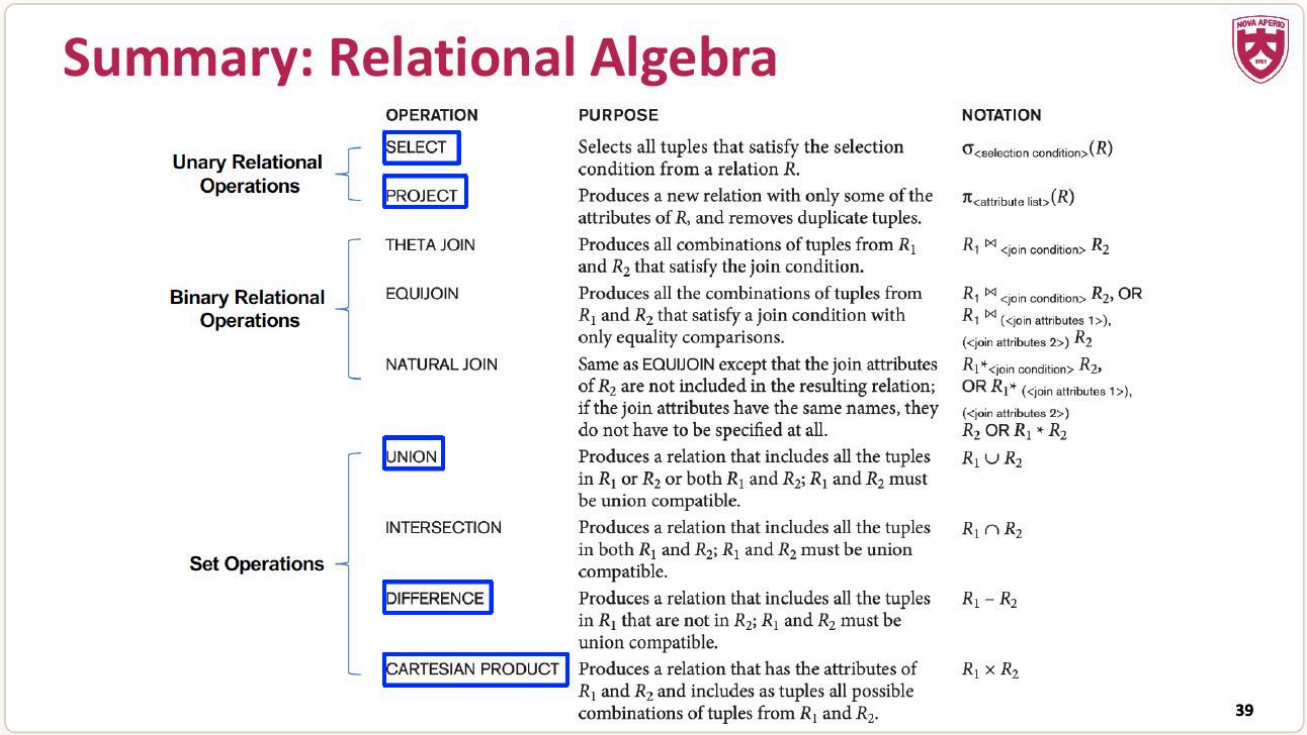


질의 평가 트리 예

- Q. Stafford 프로젝트마다 번호, 통제 부서 번호, 관리자의 성·주소·생일
  - (1) σ P.Plocation='Stafford' (P)
  - (2) (1) ⋈ P.Dnum=D.Dnumber (D)
  - (3) (2) ⋈ D.Mgr_ssn=E.Ssn (E)
  - 루트: π P.Pnumber, P.Dnum, E.Lname, E.Address, E.Bdate
- (1)이 끝나야 (2), 그다음 (3) — 질의의 좋은 시각적 표현 → 11주 질의 처리·최적화

✍ 메모

💡 이해 σ를 먼저(아래) 하면 조인할 행이 줄어든다 — 최적화의 기본 아이디어



Summary: Relational Algebra

	OPERATION	PURPOSE	NOTATION
Unary Relational Operations	SELECT	Selects all tuples that satisfy the selection condition from a relation R.	$\sigma_{\text{<selection condition>}}(R)$
	PROJECT	Produces a new relation with only some of the attributes of R, and removes duplicate tuples.	$\pi_{\text{<attribute list>}}(R)$
Binary Relational Operations	THETA JOIN	Produces all combinations of tuples from R_1 and R_2 that satisfy the join condition.	$R_1 \bowtie_{\text{<join condition>}} R_2$
	EQUIJOIN	Produces all the combinations of tuples from R_1 and R_2 that satisfy a join condition with only equality comparisons.	$R_1 \bowtie_{\text{<join condition>}} R_2$, OR $R_1 \bowtie_{\text{(<join attributes 1>), (<join attributes 2>)}} R_2$
	NATURAL JOIN	Same as EQUIJOIN except that the join attributes of R_2 are not included in the resulting relation; if the join attributes have the same names, they do not have to be specified at all.	$R_1 *_{\text{<join condition>}} R_2$, OR $R_1 *_{\text{(<join attributes 1>), (<join attributes 2>)}} R_2$ OR $R_1 * R_2$
Set Operations	UNION	Produces a relation that includes all the tuples in R_1 or R_2 or both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cup R_2$
	INTERSECTION	Produces a relation that includes all the tuples in both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cap R_2$
	DIFFERENCE	Produces a relation that includes all the tuples in R_1 that are not in R_2; R_1 and R_2 must be union compatible.	$R_1 - R_2$
	CARTESIAN PRODUCT	Produces a relation that has the attributes of R_1 and R_2 and includes as tuples all possible combinations of tuples from R_1 and R_2.	$R_1 \times R_2$

39

요약: 관계 대수

- **SELECT** σ<cond>(R) — 조건을 만족하는 튜플
- **PROJECT** π<attrs>(R) — 일부 속성, 중복 제거
- **THETA JOIN** R1 ⋈<cond> R2 — 조건을 만족하는 모든 조합
- **EQUIJOIN** — '='만 쓰는 조인
- **NATURAL JOIN** R1 * R2 — EQUIJOIN에서 R2의 조인 속성 제거
- **UNION** R1 ∪ R2 / **INTERSECTION** R1 ∩ R2 / **DIFFERENCE** R1 − R2 — union compatible
- **CARTESIAN PRODUCT** R1 × R2 — 모든 조합

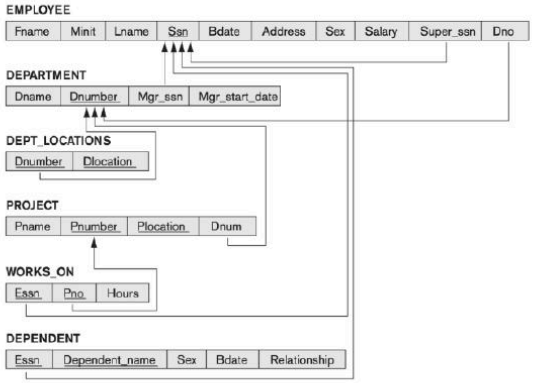
✍ 메모

★ 시험 이 표 = 기호·뜻·표기 3열로 통째로 외우기

Exercises: Relational Algebra

❑ Exercise 1: State the Following Query in Relational Algebra

■ Q. Retrieve the name and address of all employees who work for the 'Research' department



40

연습 1: 관계 대수로 쓰기

- Q. Research 부서에서 일하는 모든 직원의 이름과 주소

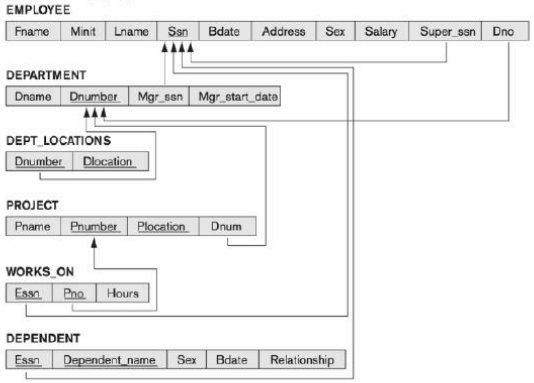
✍ 메모

✏ 예·풀이 정답은 p.61

Exercises: Relational Algebra

❑ Exercise 2: State the Following Query in Relational Algebra

■ Q. For every project located in “Stafford”, list the project number, the controlling department number, and the department manager’s last name, address, and birth date



41

연습 2: 관계 대수로 쓰기

- Q. 'Stafford'에 있는 프로젝트마다 프로젝트 번호, 통제 부서 번호, 부서 관리자의 성·주소·생일

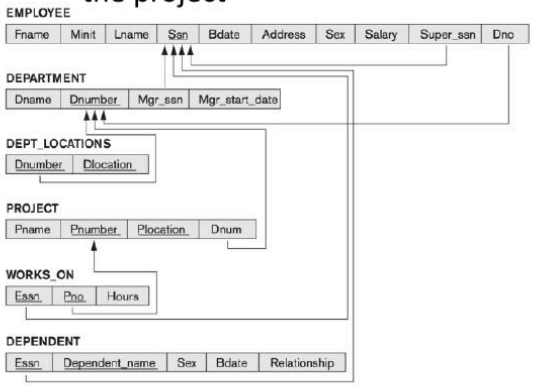
✍ 메모

✏ 예·풀이 정답은 p.62

Exercises: Relational Algebra

❑ Exercise 3: State the Following Query in Relational Algebra

■ Q. Make a list of project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project



42

연습 3: 관계 대수로 쓰기

- Q. 성이 'Smith'인 직원이 일꾼으로든, 통제 부서의 관리자로든 관련된 프로젝트 번호

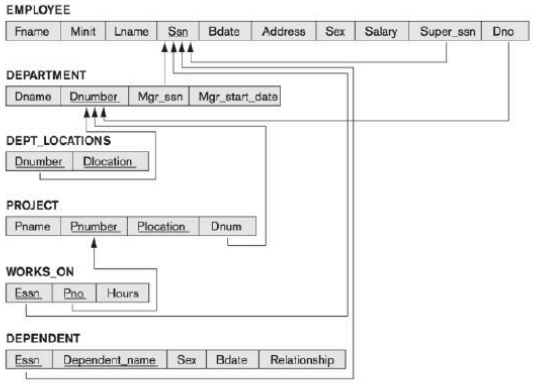
✍ 메모

✏ 예·풀이 정답은 p.63

Exercises: Relational Algebra

❑ Exercise 4: State the Following Query in Relational Algebra

■ Q. Retrieve the names of employees who have no dependents



43

연습 4: 관계 대수로 쓰기

- Q. 부양가족이 없는 직원의 이름

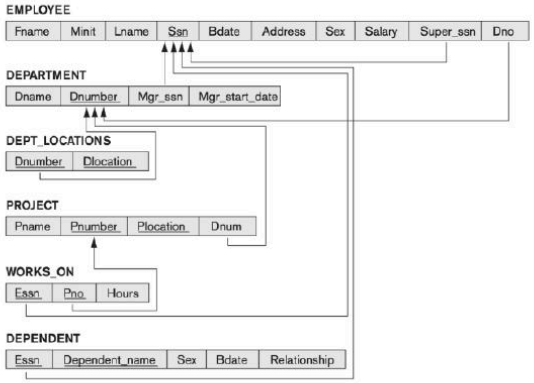
✍ 메모

✏ 예·풀이 정답은 p.64

Exercises: Relational Algebra

❑ **Exercise 5: State the Following Query in Relational Algebra**

■ Q. List the names of managers who have at least one dependent



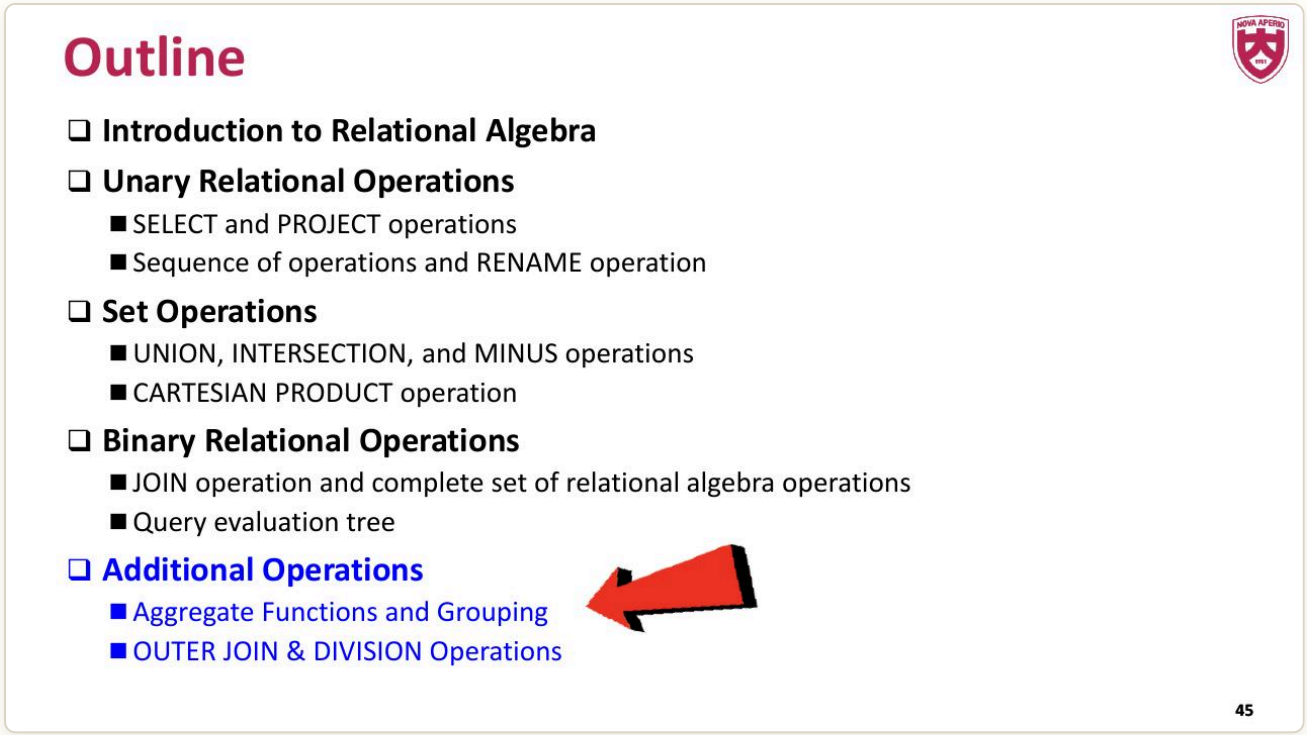
44

연습 5: 관계 대수로 쓰기

- Q. 부양가족이 한 명 이상 있는 관리자의 이름

✍ 메모

✏ 예·풀이 정답은 p.65



목차 (Outline)

- **관계 대수 소개**
- **단항 연산** — SELECT·PROJECT / 연산 잇기와 RENAME
- **집합 연산** — UNION·INTERSECTION·MINUS / CARTESIAN PRODUCT
- **이항 연산** — JOIN과 완전 집합 / 질의 평가 트리
- **추가 연산** — 집계 함수와 그룹화 / OUTER JOIN과 DIVISION

✍️ 메모

💡 이해 추가 연산

Additional Relational Operations



❑ Are the Basic Relational Algebra Operations Enough?

■ Some common DB Requests cannot be performed with the basic operations

❑ Examples

■ Requests involving *aggregate functions* on collections of values from the database

❑ To summarize information from the database tuples such as **SUM, AVERAGE, MAXIMUM**, and **COUNT**

■ Requests involving *grouping* the tuples in a relation by the value of some attributes

❑ Then, apply an aggregate function *independently to each group*

■ Requests involving *outer join operations* on two relations

Additional relational algebra operations are needed to express these requests
→ Enhancing the expressive power of the original relational algebra!

46

추가 관계 연산이 필요한 이유

- 기본 연산만으로는 흔한 DB 요청 일부를 못 한다
  - 값 모음에 대한 **집계 함수** — SUM, AVERAGE, MAXIMUM, COUNT로 요약
  - 속성 값으로 튜플을 **그룹화**하고 그룹마다 집계
  - 두 릴레이션의 **외부 조인**

✍ 메모

— 넘어가도 되는 쪽

Aggregate Functions and Grouping

❑ How to Specify Aggregate Functions in Relational Algebra?

■ Mathematical aggregate functions on collections of values from database

❑ E.g., AVERAGE, SUM, MAXIMUM, MINIMUM, and COUNT

Q1. Retrieve the average and total salary of all employees

Q2. Retrieve the total number of employees

❑ How to Specify Requests Involving Grouping the Tuples?

■ Grouping tuples by value of some attributes and applying an aggregate function

Q3. Group EMPLOYEE tuples by Dno,
so that each group includes the tuples for employees working in the same department.

Q3-1. Retrieve the average and total salary of employees in each department
Q3-2. Retrieve the number of employees in each department

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

47

집계 함수와 그룹화

- 관계 대수에서 집계 함수를 어떻게?
  - DB 값 모음에 대한 수학적 집계 — AVERAGE, SUM, MAXIMUM, MINIMUM, COUNT
- 그룹화가 들어간 요청은?
  - 속성 값으로 튜플을 묶고 그룹마다 집계 적용

✍ 메모

— 넘어가도 되는 쪽

Form of Aggregate Functions and Grouping



❑ How to Formally Define an Aggregate Function?

$$\text{<grouping attributes>}\ \mathfrak{F}\ \text{<function list>}\ (R)$$

- Symbol $\mathfrak{F}$ (Script F): the aggregate function operator
- Grouping attributes: a list of attributes of the relation *R* to be used for grouping
- Function list: a list of (<function> <attribute>) pairs
  - <function>: one of the functions such as SUM, AVERAGE, MAXIMUM, MINIMUM, and COUT
  - <attribute>: an attribute of the relation specified by *R*

❑ Output of an Aggregate Function

- A relation involving the grouping attributes with one attribute for each element in the function list

48

집계 함수의 형태 (ℱ)

```
<grouping attributes> ℱ <function list> (R)
```

- ℱ(script F): 집계 함수 연산자
- 그룹 속성: 그룹화에 쓸 R의 속성 목록
- 함수 목록: (<function> <attribute>) 쌍 목록 — SUM, AVERAGE, MAXIMUM, MINIMUM, COUNT

- 출력: **그룹 속성 + 함수마다 속성 하나**인 릴레이션

✍ 메모

★ 시험 그룹 속성은 ℱ의 왼쪽 아래첨자, 함수는 오른쪽

Aggregate Functions and Grouping: Examples

❑ Examples of AGGREGATE FUNCTION Operations

Q. Retrieve each department number, the number of employees in the department, and their average salary, while renaming the resulting attributes as indicated below

$\rho_{R(\text{Dno, No_of_employees, Average_sal})}(\text{Dno } \mathfrak{S}\ \text{COUNT Ssn, AVERAGE Salary}\ (\text{EMPLOYEE}))$.

Renaming the attributes Grouping attribute Function list

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Grouping attribute + one element for each function

R

Dno	No_of_employees	Average_sal
5	4	33250
4	3	31000
1	1	55000

Q. What if no renaming operation is specified?

Q. What if no grouping attribute is specified?

49

집계 함수 예 ①

```
ρ R(Dno, No_of_employees, Average_sal) ( Dno ℱ COUNT Ssn, AVERAGE Salary (EMPLOYEE) )
```

- → (5, 4, 33250), (4, 3, 31000), (1, 1, 55000)

- Q. 이름을 안 바꾸면? Q. 그룹 속성이 없으면?

✍ 메모

✏ 예·풀이 SQL: SELECT Dno, COUNT(*), AVG(Salary) FROM EMPLOYEE GROUP BY Dno;

Aggregate Functions and Grouping: Examples

❑ Examples of AGGREGATE FUNCTION Operations

Q. Retrieve ~~each department number~~, the number of employees ~~in the department~~, and their average salary, ~~while renaming the resulting attributes as indicated below~~

$_{Dno}\mathfrak{F}_{COUNT\ Ssn,\ AVERAGE\ Salary}(EMPLOYEE)$.

$\mathfrak{F}_{COUNT\ Ssn,\ AVERAGE\ Salary}(EMPLOYEE)$.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

The concatenation of the function name with the attribute name:
Form of <function>_<attribute>

Dno	Count_ssn	Average_salary
5	4	33250
4	3	31000
1	1	55000

Count_ssn	Average_salary
8	35125

The functions are applied to all the tuples in the relation
Thus, the resulting relation has a ***single tuple only***

50

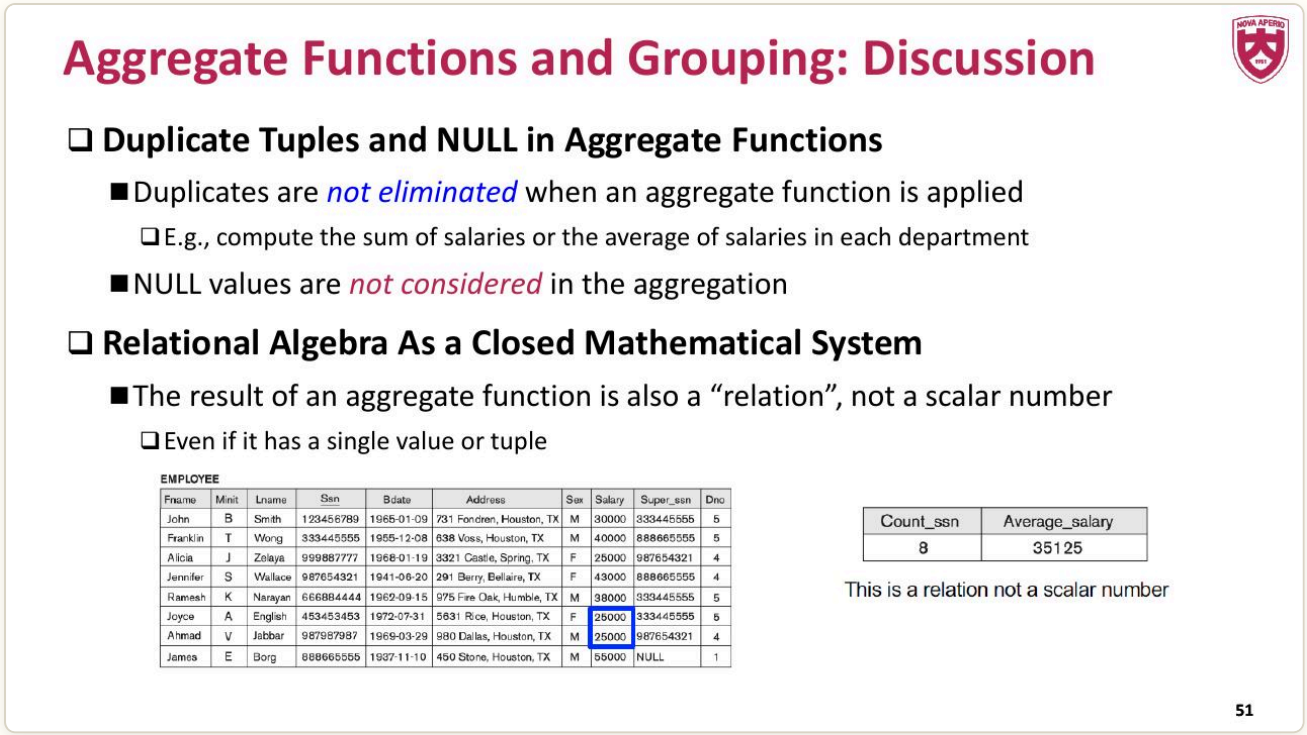
집계 함수 예 ②

- 이름을 안 바꾸면: 속성 이름이 **<function>_<attribute>** → Dno, Count_ssn, Average_salary
- 그룹 속성이 없으면: 함수를 **모든 튜플**에 적용 → 결과는 **한 튜플**

```
ℱ COUNT Ssn, AVERAGE Salary (EMPLOYEE)  → (8, 35125)
```

✍ 메모

— 넘어가도 되는 쪽



Aggregate Functions and Grouping: Discussion

❑ Duplicate Tuples and NULL in Aggregate Functions

- ■ Duplicates are *not eliminated* when an aggregate function is applied
  - ❑ E.g., compute the sum of salaries or the average of salaries in each department
- ■ NULL values are *not considered* in the aggregation

❑ Relational Algebra As a Closed Mathematical System

- ■ The result of an aggregate function is also a "relation", not a scalar number
  - ❑ Even if it has a single value or tuple

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Count_ssn	Average_salary
8	35125

This is a relation not a scalar number

51

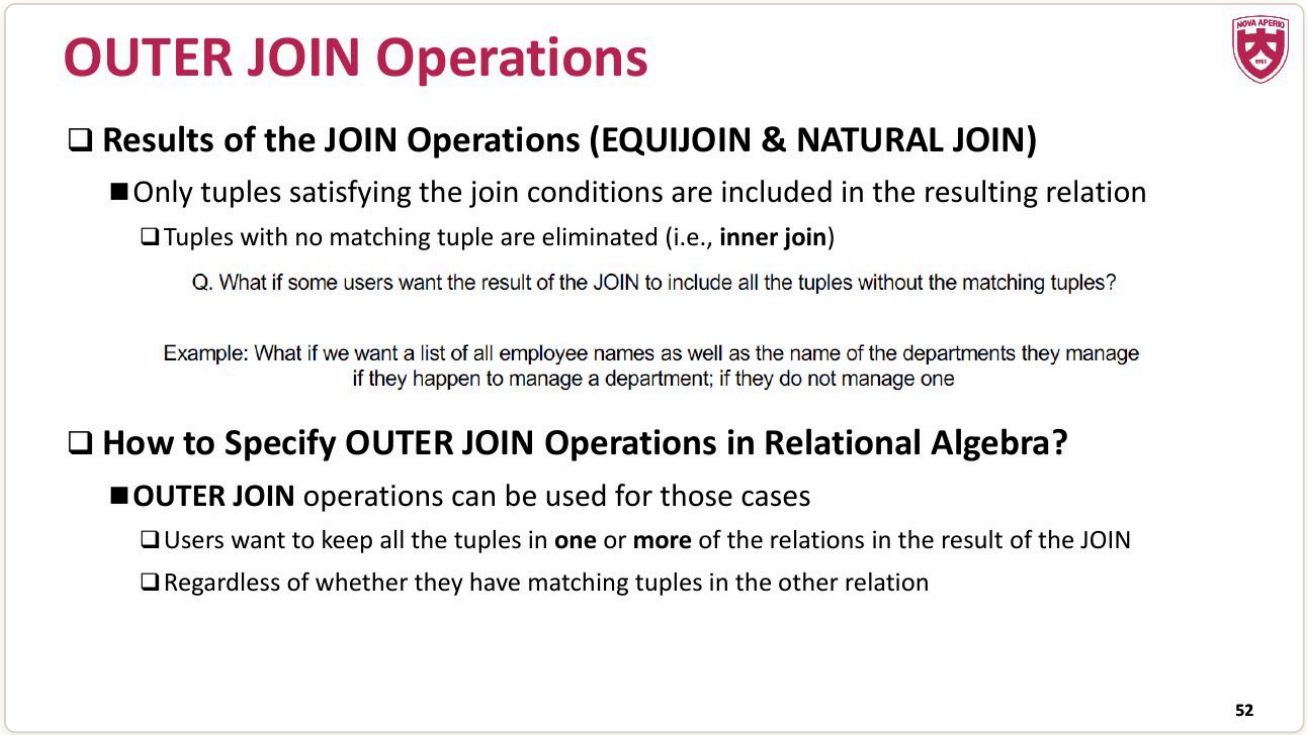
집계 함수 — 토론

- **중복은 제거하지 않는다** (부서별 월급 합·평균을 생각해 보라)
- **NULL은 고려하지 않는다**
- 관계 대수는 **닫힌 수학 체계**
  - 집계 결과도 스칼라 숫자가 아니라 **릴레이션** (값·튜플이 하나뿐이어도)

✍ 메모

⚠ 주의 π는 중복 제거, 𝔊 집계는 중복 유지 — 대비!

★ 시험 closed system: 결과가 항상 릴레이션



OUTER JOIN

- EQUIJOIN·NATURAL JOIN의 결과에는 조건을 만족하는 튜플만
  - 짝 없는 튜플은 제거 (**inner join**)
- **OUTER JOIN**은 이런 경우에
  - 한쪽(또는 양쪽) 릴레이션의 튜플을 **모두** 결과에 남기고 싶을 때
  - 다른 쪽에 짝이 있든 없든

✍ 메모

— 넘어가도 되는 쪽

OUTER JOIN Operations: Example

❑ How to Specify OUTER JOIN Operations in Relational Algebra?

Q. Retrieve all employee names as well as the name of the departments they manage if they happen to manage a department; if they do not manage one

$$TEMP \leftarrow (EMPLOYEE \overset{}{\bowtie}_{Ssn=Mgr_ssn} DEPARTMENT)$$
$$RESULT \leftarrow \pi_{Fname,\ Minit,\ Lname,\ Dname}(TEMP)$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

RESULT

Fname	Minit	Lname	Dname
John	B	Smith	NULL
Franklin	T	Wong	Research
Alicia	J	Zelaya	NULL
Jennifer	S	Wallace	Administration
Ramesh	K	Narayan	NULL
Joyce	A	English	NULL
Ahmad	V	Jabbar	NULL
James	E	Borg	Headquarters

Padded with NULL values

Including every tuple in the first (**left**) relation

❑ Types of OUTER JOIN Operations

⟕ LEFT OUTER JOIN ⟖ RIGHT OUTER JOIN ⟗ FULL OUTER JOIN

53

OUTER JOIN 예와 종류

- Q. 모든 직원의 이름과, 부서를 관리하면 그 부서 이름

```
TEMP ← EMPLOYEE ⟕ Ssn=Mgr_ssn DEPARTMENT
RESULT ← π Fname, Minit, Lname, Dname (TEMP)
```

  - → Wong–Research, Wallace–Administration, Borg–Headquarters, 나머지 직원은 Dname **NULL**
- 종류: **LEFT ⟕ / RIGHT ⟖ / FULL ⟗ OUTER JOIN**

✍ 메모

★ 시험 기호 방향 = 다 남기는 쪽 (⟕ 왼쪽 전부)

DIVISION Operation

❑ What is the DIVISION Operation?

■ Useful for a special kind of query that sometimes occurs in DB applications

❑ This operation, denoted by ÷, can be performed with the basic relational algebra operations

■ Example:

❑ Retrieve the names of employees who work on ***all the projects*** that 'John Smith' works on

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

The projects 1 &2

54

DIVISION (÷)

- DB 응용에서 가끔 나오는 **특별한 질의**에 유용
  - ÷로 표기 — 기본 관계 대수 연산으로도 표현 가능
- 예: 'John Smith'가 일하는 **모든** 프로젝트에서 일하는 직원 이름

✍ 메모

★ 시험 질문에 '모든(all)'이 보이면 DIVISION

DIVISION Operation: Example

❑ How to Specify the DIVISION Operation in Relational Algebra?

Q.Retrieve the names of employees who work on ***all*** the projects that 'John Smith' works on

Find all the projects that 'John Smith' works on

$SMITH \leftarrow \sigma_{Fname='John' \text{ AND } Lname='Smith'}(EMPLOYEE)$

$SMITH_PNOS \leftarrow \pi_{Pno}(WORKS_ON \bowtie_{Essn=Ssn} SMITH)$

Create a relation about employees working on projects

$SSN_PNOS \leftarrow \pi_{Essn, Pno}(WORKS_ON)$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

SMITH_PNOS

Pno
1
2

SSN_PNOS

Essn	Pno
123456789	1
123456789	2
666884444	3
453453453	1
453453453	2
333445555	2
333445555	3
333445555	10
333445555	20
999887777	30
999887777	10
987987987	10
987987987	30
987654321	30
987654321	20
888665555	20

<Pno, Essn> Relation

55

DIVISION 예 ①

```
SMITH ← σ Fname='John' AND Lname='Smith' (EMPLOYEE)
SMITH_PNOS ← π Pno (WORKS_ON ⋈ Essn=Ssn SMITH)   → {1, 2}
SSN_PNOS ← π Essn, Pno (WORKS_ON)   – <Essn, Pno> 쌍
```

✍ 메모

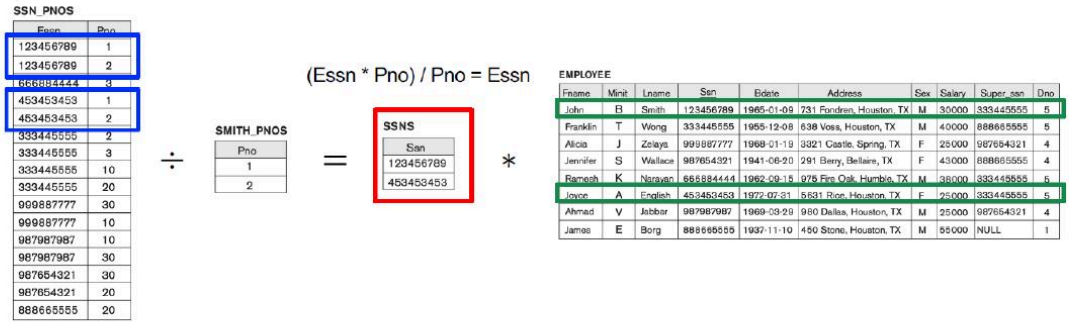
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DIVISION Operation: Example (Cont.)

❑ How to Specify the DIVISION Operation in Relational Algebra?

Q.Retrieve the names of employees who work on ***all*** the projects that 'John Smith' works on

SSNS(Ssn) ← SSN_PNOS ÷ SMITH_PNOS
RESULT ← $\pi_{Fname, Lname}$(SSNS * EMPLOYEE)



By applying the DIVISION operation, the desired SSNs are retrieved

56

DIVISION 예 ②

```
SSNS(Ssn) ← SSN_PNOS ÷ SMITH_PNOS   → {123456789, 453453453}
```

- (Essn, Pno) ÷ (Pno) = (Essn)

```
RESULT ← π Fname, Lname (SSNS * EMPLOYEE)   → John Smith, Joyce English
```

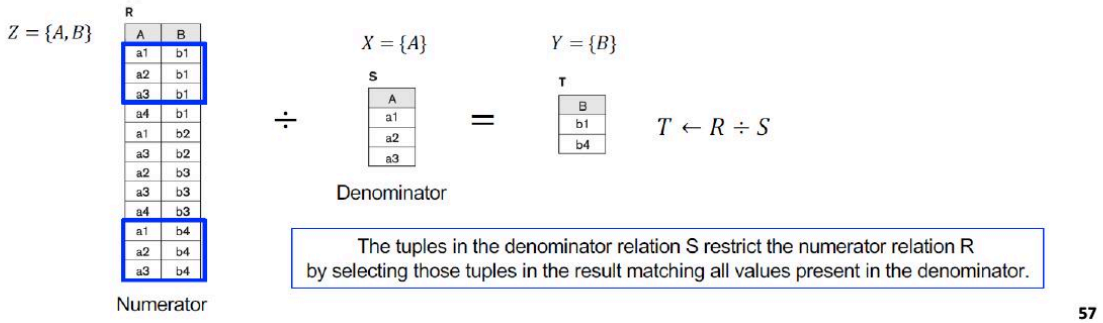
✍ 메모

✏ 예·풀이 Pno 1과 2를 둘 다 가진 Essn만 남음 — Smith 자신과 English

DIVISION Operation: Usage

❑ Form of the DIVISION Operation

- ■ The DIVISION operation is applied to two relations *R(Z)* and *S(X)*
  - ❑ $R(Z) \div S(X)$, $X \subseteq Z$ (the attributes of S are a subset of the attributes of R)
- ■ The result of this operation is a relation *T(Y)*
  - ❑ $Y = Z - X$ (i. e. , $X \cup Y = Z$)
  - ❑ I.e., it includes a tuple t_R if tuples t_R appear in R with $t_R[Y] = t_S$, and with $t_R[X] = t_S$ for every tuple t_S



57

DIVISION의 형태

`R(Z) ÷ S(X),  X ⊆ Z  (S의 속성은 R 속성의 부분집합)`

- 결과 T(Y), **Y = Z − X** (X ∪ Y = Z)
  - tR[Y]가 결과에 들어가려면: S의 **모든** 튜플 tS에 대해 tR[X] = tS인 튜플이 R에 있어야
- 예: R(A, B)와 S(A) = {a1, a2, a3} → T(B) = **{b1, b4}**
  - 분모 S의 모든 값과 짝이 맞는 것만 결과에

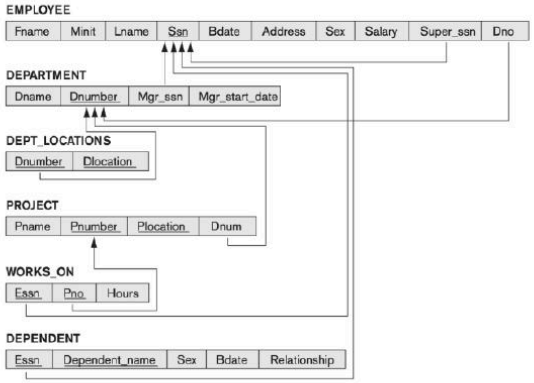
✍ 메모

✏ 예·풀이 b1은 a1·a2·a3(+a4) 모두와 짝, b4도 a1·a2·a3과 짝
→ 통과. b2(a1,a3), b3(a2,a3,a4)는 a1 또는 a2가 없음

Exercises: Additional Relational Algebra

❑ **Exercise 6: State the Following Query in Relational Algebra**

■ Q. List the names of all employees with two or more dependents



58

연습 6: 추가 관계 대수

- Q. 부양가족이 **2명 이상**인 모든 직원의 이름

✍ 메모

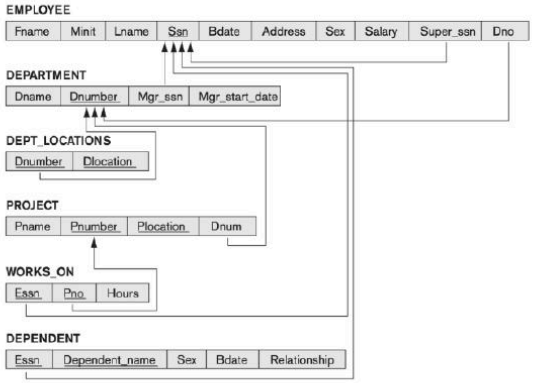
✏ 예·풀이 정답 p.66 — ❀ COUNT 다음 σ

Exercises: Additional Relational Algebra



❑ Exercise 7: State the Following Query in Relational Algebra

■ Q. Find the names of employees who work on *all* the projects controlled by department number 5



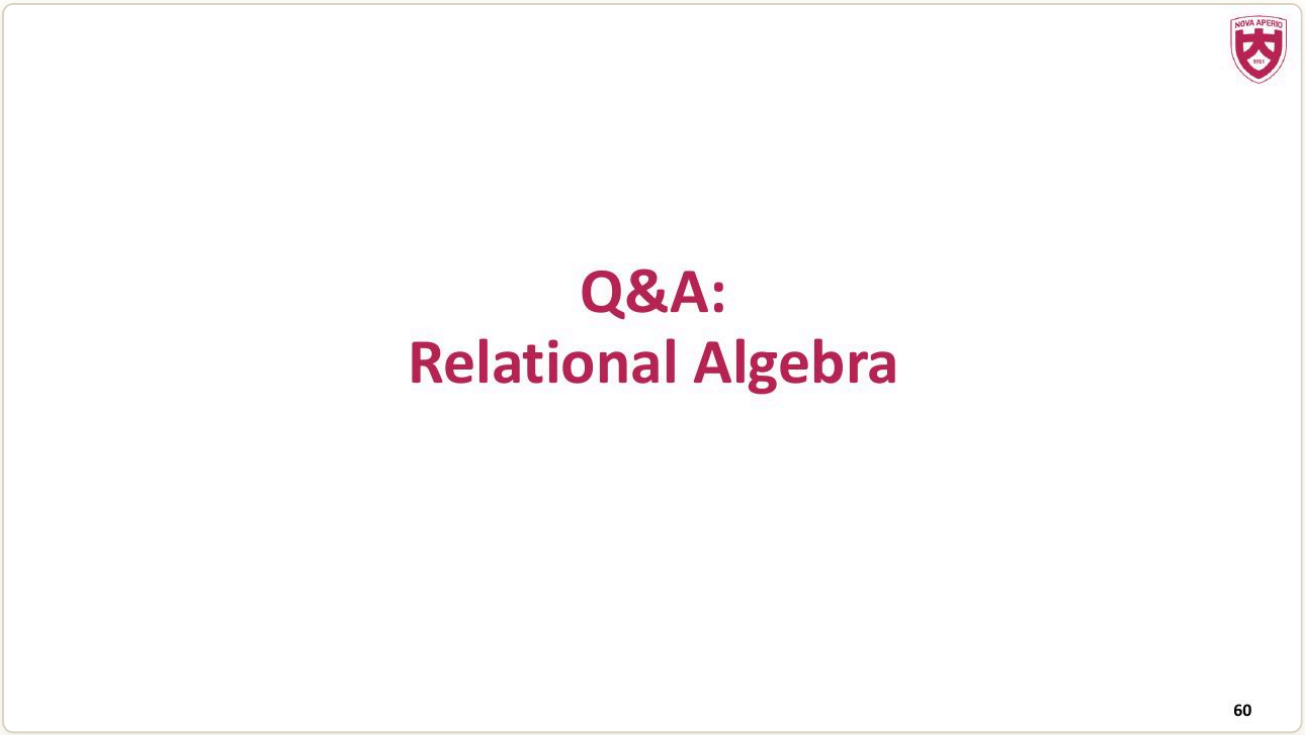
59

연습 7: 추가 관계 대수

- Q. 5번 부서가 통제하는 **모든** 프로젝트에서 일하는 직원의 이름

✍ 메모

✏ 예·풀이 정답 p.67 — DIVISION



Q&A

- Relational Algebra — 끝
- p.61~67은 연습문제 정답

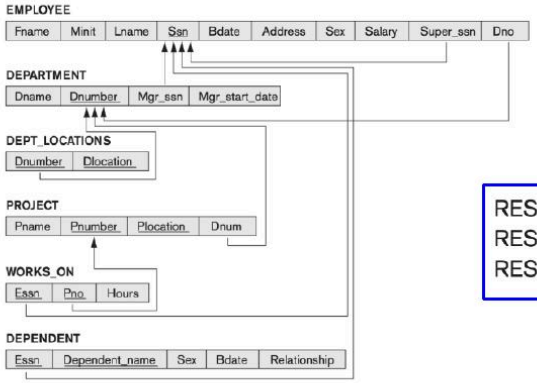
✍ 메모

— 넘어가도 되는 쪽

Exercises: Relational Algebra

❑ Exercise 1: State the Following Query in Relational Algebra

■ Q. Retrieve the name and address of all employees who work for the 'Research' department



$$RESEARCH_DEPT \leftarrow \sigma_{Dname='Research'}(DEPARTMENT)$$
$$RESEARCH_EMPS \leftarrow (RESEARCH_DEPT \bowtie_{Dnumber=Dno} EMPLOYEE)$$
$$RESULT \leftarrow \pi_{Fname,\ Lname,\ Address}(RESEARCH_EMPS)$$

61

연습 1 정답: 관계 대수로 쓰기

- Q. Research 부서에서 일하는 모든 직원의 이름과 주소

```
RESEARCH_DEPT ← σ Dname='Research' (DEPARTMENT)
RESEARCH_EMPS ← RESEARCH_DEPT ⋈ Dnumber=Dno EMPLOYEE
RESULT ← π Fname, Lname, Address (RESEARCH_EMPS)
```

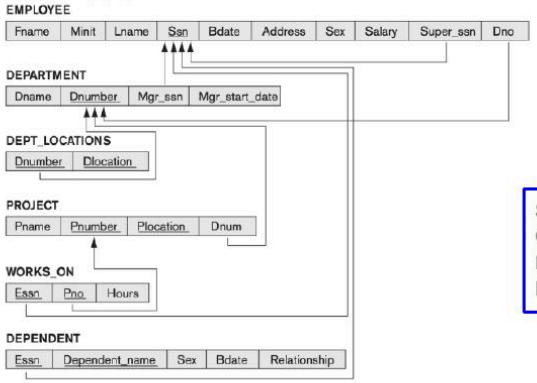
✍ 메모

★ 시험 패턴: σ(고르기) → ⋈(잇기) → π(필요한 열만)

Exercises: Relational Algebra

❑ Exercise 2: State the Following Query in Relational Algebra

■ Q. For every project located in “Stafford”, list the project number, the controlling department number, and the department manager’s last name, address, and birth date



$$\text{STAFFORD_PROJS} \leftarrow \sigma_{\text{Plocation='Stafford'}}(\text{PROJECT})$$
$$\text{CONTR_DEPTS} \leftarrow (\text{STAFFORD_PROJS} \bowtie_{\text{Dnum=Dnumber}} \text{DEPARTMENT})$$
$$\text{PROJ_DEPT_MGRS} \leftarrow (\text{CONTR_DEPTS} \bowtie_{\text{Mgr_ssn=Ssn}} \text{EMPLOYEE})$$
$$\text{RESULT} \leftarrow \pi_{\text{Pnumber, Dnum, Lname, Address, Bdate}}(\text{PROJ_DEPT_MGRS})$$

62

연습 2 정답: 관계 대수로 쓰기

- Q. 'Stafford'에 있는 프로젝트마다 프로젝트 번호, 통제 부서 번호, 부서 관리자의 성·주소·생일

```
STAFFORD_PROJS ← σ Plocation='Stafford' (PROJECT)
CONTR_DEPTS ← STAFFORD_PROJS ⋈ Dnum=Dnumber DEPARTMENT
PROJ_DEPT_MGRS ← CONTR_DEPTS ⋈ Mgr_ssn=Ssn EMPLOYEE
RESULT ← π Pnumber, Dnum, Lname, Address, Bdate (PROJ_DEPT_MGRS)
```

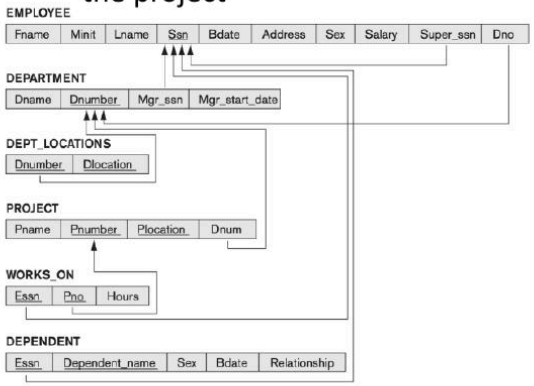
✍ 메모

💡 이해 조인 두 번 — 질의 평가 트리(p.38)와 같은 문제

Exercises: Relational Algebra

❑ Exercise 3: State the Following Query in Relational Algebra

■ Q. Make a list of project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project



Get the project numbers as a worker

$SMITHS(Essn) \leftarrow \pi_{Ssn}(\sigma_{Lname='Smith'}(EMPLOYEE))$
$SMITH_WORKER_PROJS \leftarrow \pi_{Pno}(WORKS_ON * SMITHS)$

Get the project numbers as a manager

$MGRS \leftarrow \pi_{Lname, Dnumber}(EMPLOYEE \bowtie_{Ssn=Mgr_ssn} DEPARTMENT)$
$SMITH_MANAGED_DEPTS(Dnum) \leftarrow \pi_{Dnumber}(\sigma_{Lname='Smith'}(MGRS))$
$SMITH_MGR_PROJS(Pno) \leftarrow \pi_{Pnumber}(SMITH_MANAGED_DEPTS * PROJECT)$

Combine the results

$RESULT \leftarrow (SMITH_WORKER_PROJS \cup SMITH_MGR_PROJS)$

63

연습 3 정답: 관계 대수로 쓰기

- Q. 성이 'Smith'인 직원이 일꾼으로든, 통제 부서의 관리자로든 관련된 프로젝트 번호

```
SMITHS(Essn) ← π Ssn (σ Lname='Smith' (EMPLOYEE))
SMITH_WORKER_PROJS ← π Pno (WORKS_ON * SMITHS)
MGRS ← π Lname, Dnumber (EMPLOYEE ⋈ Ssn=Mgr_ssn DEPARTMENT)
SMITH_MANAGED_DEPTS(Dnum) ← π Dnumber (σ Lname='Smith' (MGRS))
SMITH_MGR_PROJS(Pno) ← π Pnumber (SMITH_MANAGED_DEPTS * PROJECT)
RESULT ← SMITH_WORKER_PROJS ∪ SMITH_MGR_PROJS
```

✍ 메모

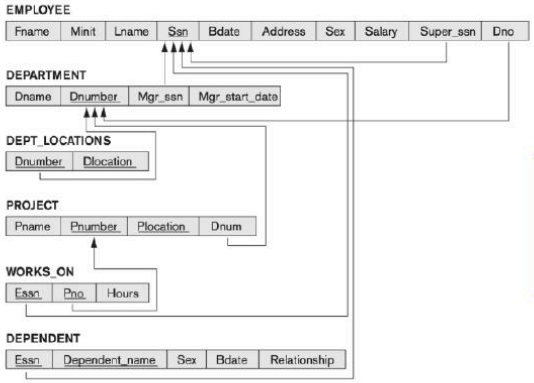
★ 시험 '~이거나' = ∪ (두 경로 결과를 합침)

⚠ 주의 ∪ 하려면 두 결과가 type-compatible이어야 → ρ로 속성 이름을 Pno로 맞춤

Exercises: Relational Algebra

❑ Exercise 4: State the Following Query in Relational Algebra

■ Q. Retrieve the names of employees who have no dependents



$$ALL_EMPS \leftarrow \pi_{Ssn}(EMPLOYEE)$$
$$EMPS_WITH_DEPS(Ssn) \leftarrow \pi_{Essn}(DEPENDENT)$$
$$EMPS_WITHOUT_DEPS \leftarrow (ALL_EMPS - EMPS_WITH_DEPS)$$
$$RESULT \leftarrow \pi_{Lname,\ Fname}(EMPS_WITHOUT_DEPS * EMPLOYEE)$$

64

연습 4 정답: 관계 대수로 쓰기

- Q. 부양가족이 없는 직원의 이름

```
ALL_EMPS ← π Ssn (EMPLOYEE)
EMPS_WITH_DEPS(Ssn) ← π Essn (DEPENDENT)
EMPS_WITHOUT_DEPS ← ALL_EMPS – EMPS_WITH_DEPS
RESULT ← π Lname, Fname (EMPS_WITHOUT_DEPS * EMPLOYEE)
```

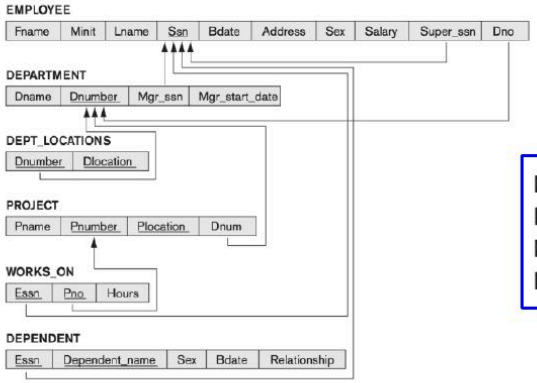
✍ 메모

★ 시험 '없는' = 전체 – 있는 것 (MINUS)

Exercises: Relational Algebra

❑ Exercise 5: State the Following Query in Relational Algebra

■ Q. List the names of managers who have at least one dependent



$$MGRS(Ssn) \leftarrow \pi_{Mgr_ssn}(DEPARTMENT)$$
$$EMPS_WITH_DEPS(Ssn) \leftarrow \pi_{Essn}(DEPENDENT)$$
$$MGRS_WITH_DEPS \leftarrow (MGRS \cap EMPS_WITH_DEPS)$$
$$RESULT \leftarrow \pi_{Lname,\ Fname}(MGRS_WITH_DEPS * EMPLOYEE)$$

65

연습 5 정답: 관계 대수로 쓰기

- Q. 부양가족이 한 명 이상 있는 관리자의 이름

```
MGRS(Ssn) ← π Mgr_ssn (DEPARTMENT)
EMPS_WITH_DEPS(Ssn) ← π Essn (DEPENDENT)
MGRS_WITH_DEPS ← MGRS ∩ EMPS_WITH_DEPS
RESULT ← π Lname, Fname (MGRS_WITH_DEPS * EMPLOYEE)
```

✍ 메모

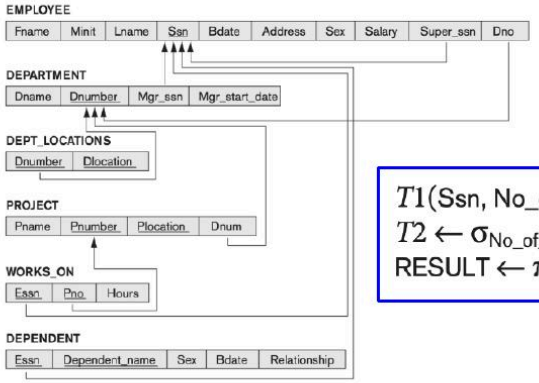
★ 시험 '~이면서 ~인' = ∩

✏ 예·풀이 결과: Wong, Wallace

Exercises: Additional Relational Algebra

❑ **Exercise 6: State the Following Query in Relational Algebra**

■ Q. List the names of all employees with two or more dependents



$T1(\text{Ssn, No_of_dependents}) \leftarrow {}_{\text{Essn}}\Im_{\text{COUNT Dependent_name}}(\text{DEPENDENT})$

$T2 \leftarrow \sigma_{\text{No_of_dependents}>2}(T1)$

$\text{RESULT} \leftarrow \pi_{\text{Lname, Fname}}(T2 * \text{EMPLOYEE})$

66

연습 6 정답: 관계 대수로 쓰기

- Q. 부양가족이 2명 이상인 모든 직원의 이름

```
T1(Ssn, No_of_dependents) ← Essn ℑ COUNT Dependent_name (DEPENDENT)
T2 ← σ No_of_dependents > 2 (T1)
RESULT ← π Lname, Fname (T2 * EMPLOYEE)
```

✍ 메모

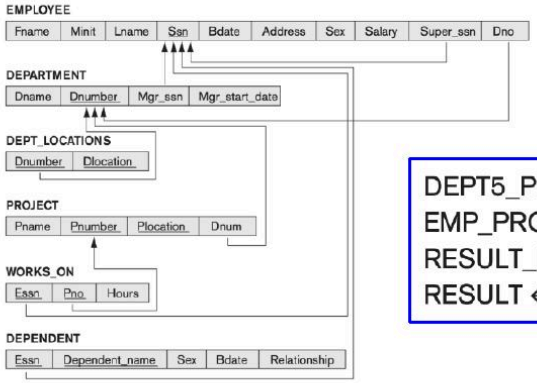
⚠ 주의 교안은 '> 2'로 썼지만 'two or more'면 ≥ 2가 맞다 — 시험엔 ≥ 2로

★ 시험 그룹 집계 ℑ 다음 σ — SQL의 GROUP BY + HAVING에 해당

Exercises: Additional Relational Algebra

❑ Exercise 7: State the Following Query in Relational Algebra

■ Q. Find the names of employees who work on *all* the projects controlled by department number 5



$$DEPT5_PROJS \leftarrow \rho_{(Pno)}(\pi_{Pnumber}(\sigma_{Dnum=5}(PROJECT)))$$
$$EMP_PROJ \leftarrow \rho_{(Ssn,\ Pno)}(\pi_{Essn,\ Pno}(WORKS_ON))$$
$$RESULT_EMP_SSNS \leftarrow EMP_PROJ \div DEPT5_PROJS$$
$$RESULT \leftarrow \pi_{Lname,\ Fname}(RESULT_EMP_SSNS * EMPLOYEE)$$

67

연습 7 정답: 관계 대수로 쓰기

- Q. 5번 부서가 통제하는 모든 프로젝트에서 일하는 직원의 이름

```
DEPT5_PROJS ← ρ(Pno) (π Pnumber (σ Dnum=5 (PROJECT)))
EMP_PROJ ← ρ(Ssn, Pno) (π Essn, Pno (WORKS_ON))
RESULT_EMP_SSNS ← EMP_PROJ ÷ DEPT5_PROJS
RESULT ← π Lname, Fname (RESULT_EMP_SSNS * EMPLOYEE)
```

✍ 메모

★ 시험 '모든 ~에' = DIVISION ÷ (4강 연습 4의 NOT EXISTS ... EXCEPT와 같은 뜻)